WEEKLY ROLLUP • PhD RESEARCH WATCH

Particulate Matter
Week in Review

Week ending 10 October 2026 | 4 October 2026 to 10 October 2026 | 87 records pooled from 7 daily issues

87
RECORDS POOLED
(87 UNIQUE DOIs)

44%
MEASUREMENT SIDE
(38 RECORDS)

10
SUBTOPICS
ACTIVE

16
POOLED EFFECT
ESTIMATES (16 WITH A
CI)

27
ABSTRACT-FREE
(31% OF WEEK)

The week in one paragraph

Volume fell another 30 % to 87 records, the measurement side rose to 44 %, and the abstract gap widened to 31 % — its worst reading of the series. The week's two most useful records both say the same thing: the number a network reports is not the quantity a health study needs.

Scale. 87 records over seven issues on daily totals of **5**, 13, 11, 14, 16, **19** and 9 — a 3.8× range, much flatter than last week's 7.5×. Against recent rollups: 87 after 125, 162, 148, 126 and 150. This is the smallest week in the series. **All ten subtopics are populated** for the third week running, though two of them hold a single record.

The measurement side rose to 44 %. *Exposure assessment & modelling* 32 → **23** and *Sensing* 20 → **15** both fell in absolute terms, but less than the week did, so the share reads 40, 44, 33, 42, **44 %** across five weeks. The recovery noted last week holds.

Two records worth the week. A **preregistered external validation** of spatially valid conformal prediction for exposure mapping (8 October) reached **0.873** coverage at nominal 90 % under spatial hold-out against **0.486** for gradient boosting — then fell to **0.794** under temporal hold-out and **failed its preregistered joint hypothesis**. And a critical review of **66** low-cost-sensor calibration studies (10 October) puts median calibrated R^2 at **0.92** in the laboratory against **0.72** in the field, with median RMSE rising from **2.6** to **12.3** $\mu\mathbf{g\,m^{-3}}$. One says the map's uncertainty interval is not honest out of sample; the other says the node's calibration does not transfer. They are the same finding at two scales.

The abstract gap got worse, not better. 27 of 87 (31 %), against 22 % last week — the highest share of the series — and **14 of the 27** are measurement-side records. The **Scholar Gateway** connector, the designated route for recovering numbers that abstracts omit, has now failed on **every attempt for two full weeks** with `ACCESS_DENIED`. It needs reconnecting at the user's end; until it is, the retry chain is PubMed, Europe PMC and Semantic Scholar only, and it recovers nothing from Elsevier and Springer same-day deposits.

Estimates: 16, every one with an interval. One estimate per 5.4 records (series 2.7, 3.1, 5.2, 4.2, **5.4**), **16 of 16 with a CI** — the third complete week in a row. Two issues (4 and 10 October) contributed none.

Subtopic trend and inflection

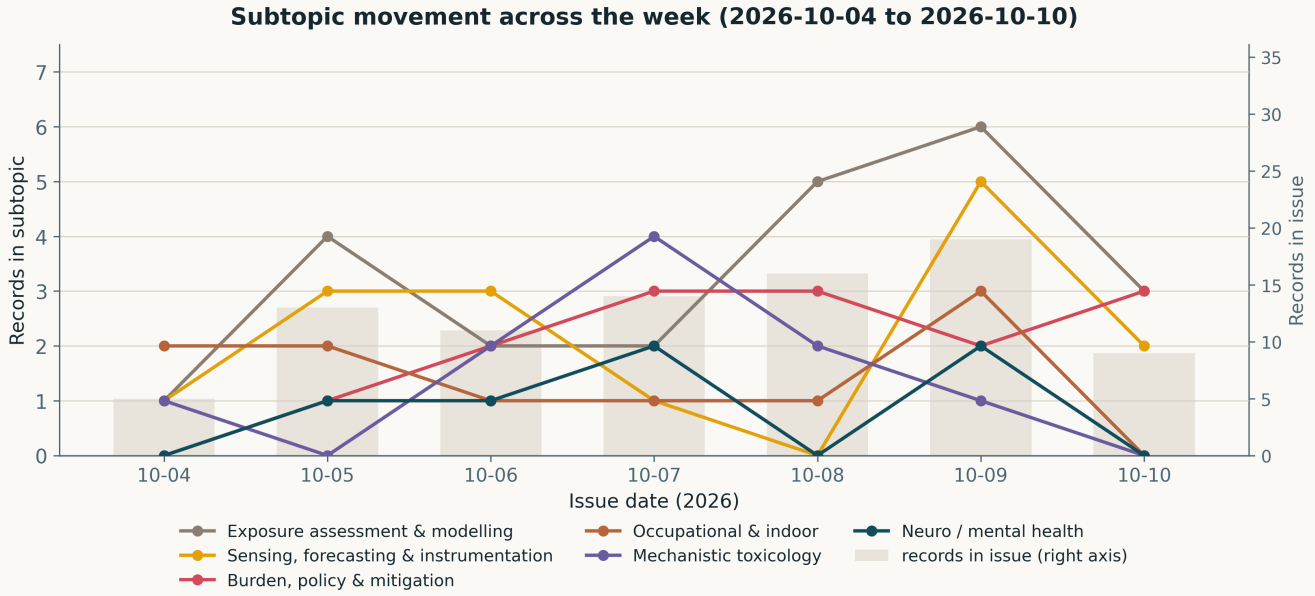


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`, seven issues. Daily totals 5, 13, 11, 14, 16, 19 and 9. Six of the seven issues were authored on the **9 October recovery run** or later, and the last two (9 and 10 October) on the **12 October run** — the period was reconstructed rather than observed day by day, which is why the day-to-day profile is flatter than usual: a single screening pass across several cached windows applies one standard where seven separate runs would have drifted.

Expanding. *Burden, policy & mitigation* holds at **14** on a week 30 % smaller, the strongest relative gain: the tyre-and-road-wear inventory, the Chinese black-carbon burden revision, the NYC congestion-pricing evaluation, the Kazakhstan code reform and the China–US lung-cancer projection. *Mechanistic toxicology* holds at **10**.

Contracting. *Cardiovascular & metabolic* **8** → **1** and *Reproductive & developmental* **8** → **1** are the collapses of the week; both had led the health side within the last month. *Respiratory & allergic* **5** → **3** sets a new series low for the second consecutive week. Part of the reproductive fall is an artefact of this run: a congenital-heart-defect proteomics case-control admitted to 9 October is **held, not rejected**, and is absent from this pool.

The inflection worth naming. The constituent question moved from mass to number, and the measurement gap widened with it. The 5 October record — 2.1 million adults, **dementia mortality HR 1.22** (1.17–1.27) per 10,000 cm^{-3} of ultrafine particle number — is an exposure no optical node in this corpus can see, on an endpoint with no mass-based equivalent in the week. Beside it, the 8 October source-resolved deposition model gives a deposited fraction ranging from **76 %** (new particle formation) to **27 %** (solid fuel) at the same ambient number concentration, and the 9 October adductomics record puts **39 %** of the toxicity of internally exposed $PM_{2.5}$ organics on **reduced organic nitrogen** compounds against 23 % for oxygenated organics. Three independent lines, one conclusion: **mass is the wrong exposure metric and the field now has specific alternatives** — number below 100 nm, source-resolved deposited dose, and nitrogen-containing organic composition. Of those, only the last is within reach of a cheap instrument, through multi-wavelength absorption, since the same week's North China Plain work ties brown-carbon absorptivity to the N:C ratio. **The test for next week**: whether any non-Chinese cohort reports a number-concentration or constituent-resolved estimate, and whether any sensor paper reports a validated sub-100 nm number channel.

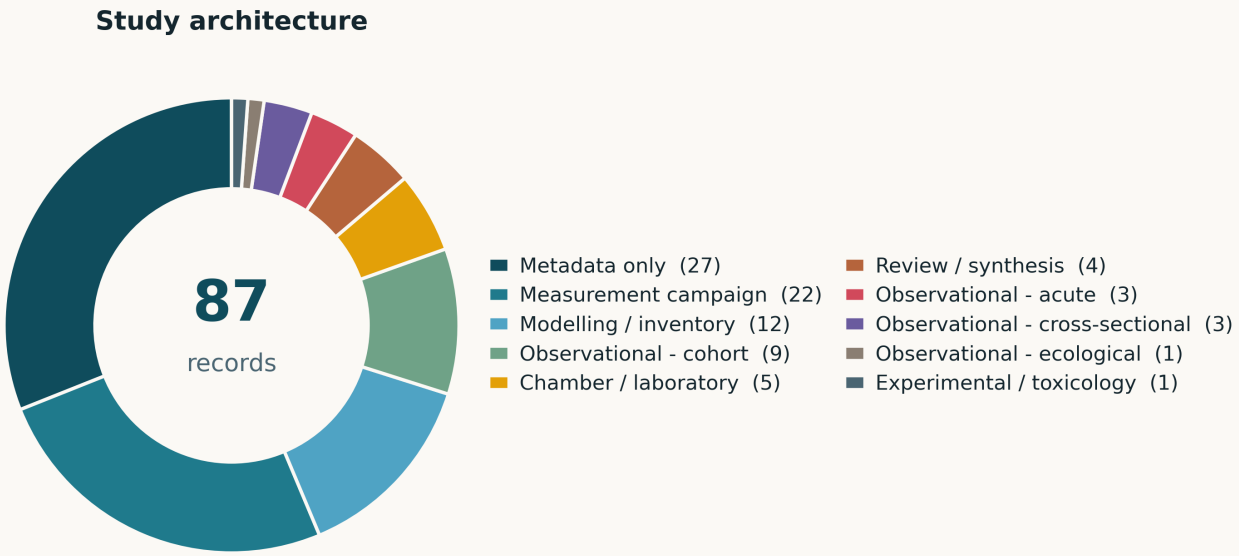


Fig. Study architecture over the pooled week. ***Metadata only* leads for the third consecutive week at 27**, now **31 %** of all records, ahead of *Measurement campaign* (22) and *Modelling / inventory* (12). Primary observational epidemiology (cohort 9, acute 3, cross-sectional 3, ecological 1) totals **16**, down from 30 and the weakest of the series. **Again no interventional record entered.**

Pooled analytics

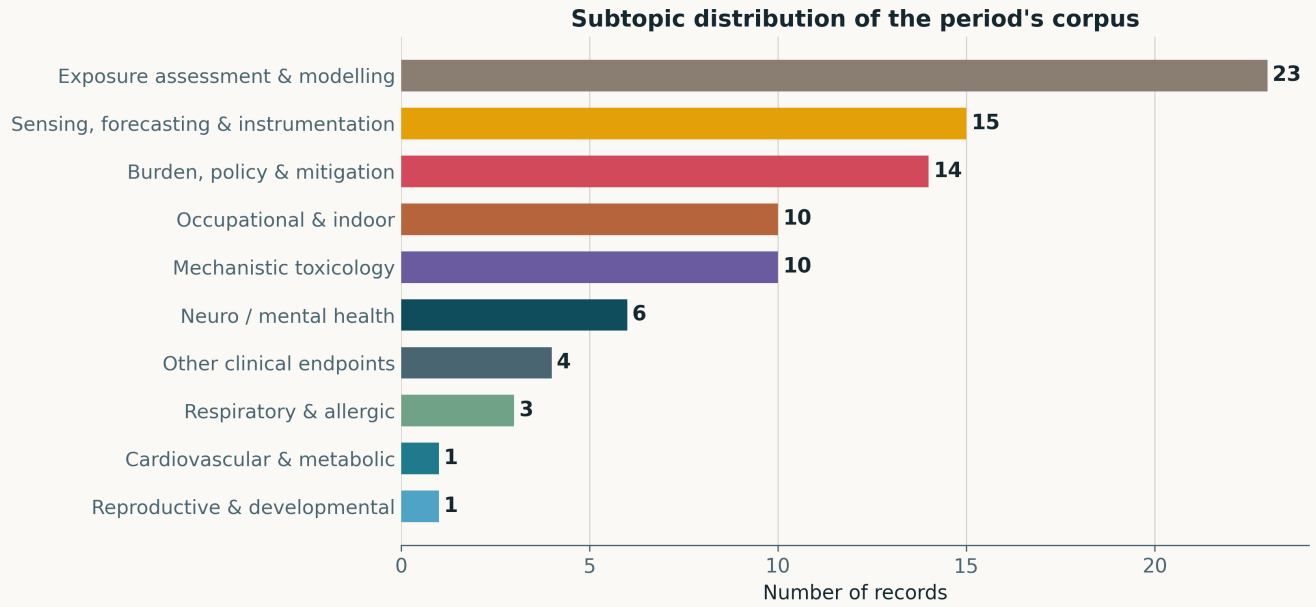


Fig. Pooled subtopic distribution, **all ten bands populated** but thinly: *Exposure* 23 and *Sensing* 15 put the measurement side at **38 of 87 (44 %)**, while *Cardiovascular & metabolic* and *Reproductive & developmental* hold **one record each**. On the health side *Burden* 14 leads, then *Occupational & indoor* and *Mechanistic toxicology* at 10.

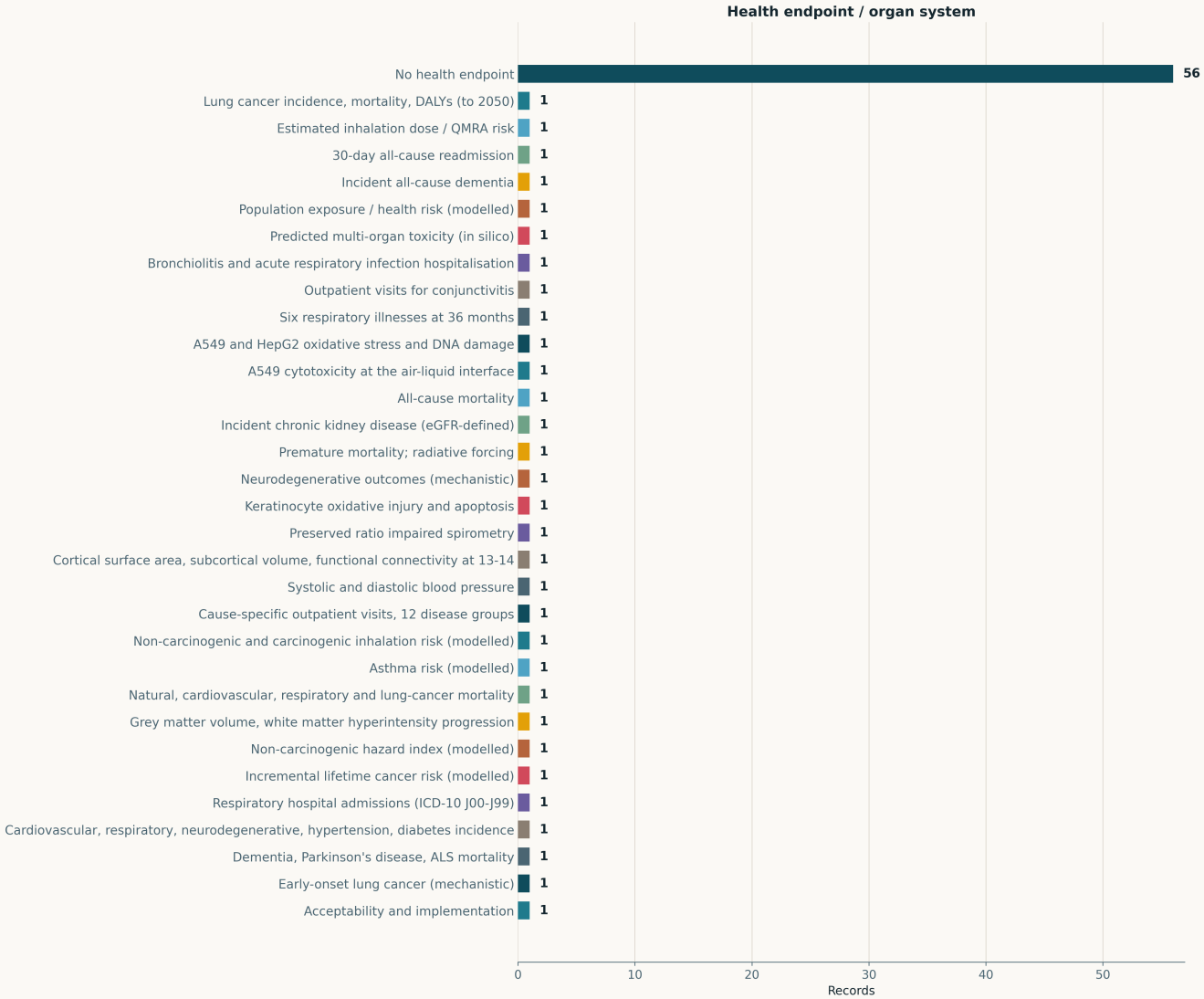


Fig. Health endpoint over the pooled week, canonicalised. **"No health endpoint" again pools the majority** — the 27 abstract-free records plus the exposure, instrument, emissions and optics work. Among human outcomes, mortality and neurological outcomes lead; no outcome label reaches five for the second consecutive week, which is what a measurement-weighted week looks like.

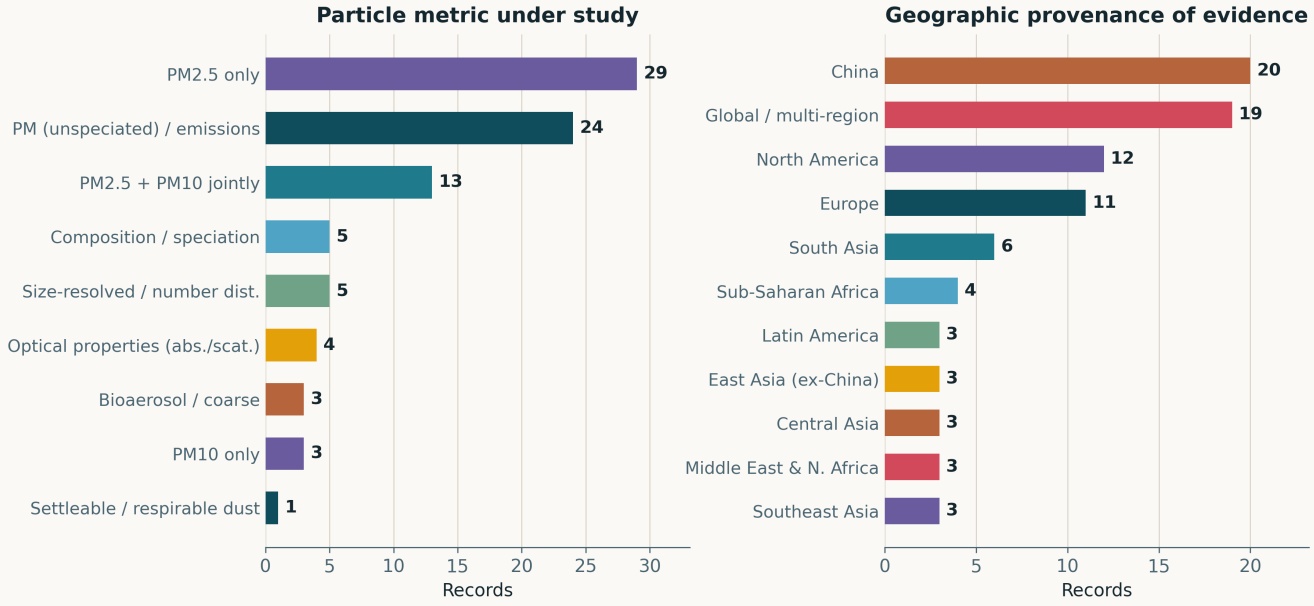


Fig. Left — particle metric. **$PM_{2.5}$ alone 29 of 87**, unspeciated PM or emissions **24**, joint $PM_{2.5}$ + PM_{10} 13. The tail is where the week's argument sits: **size-resolved number 5** and **composition / speciation 5** — the UFP dementia cohort, the Budapest deposition model, the fast-scan MPSS calibration, the dormitory Stokes-number record and the Belgrade and Pretoria speciation work — plus optical properties 4, PM_{10} alone 3, bioaerosol 3 and respirable dust 1. Right — geography. **China 20** is the largest single-country bar (23 %), then *global / multi-region* **19**, which again overstates reach: it pools reviews, laboratory work and the **13 metadata-only records whose country is not recoverable**. North America 12, Europe 11, South Asia 6, sub-Saharan Africa 4, and three each for Latin America, Central Asia, East Asia (ex-China), the Middle East & North Africa and Southeast Asia.

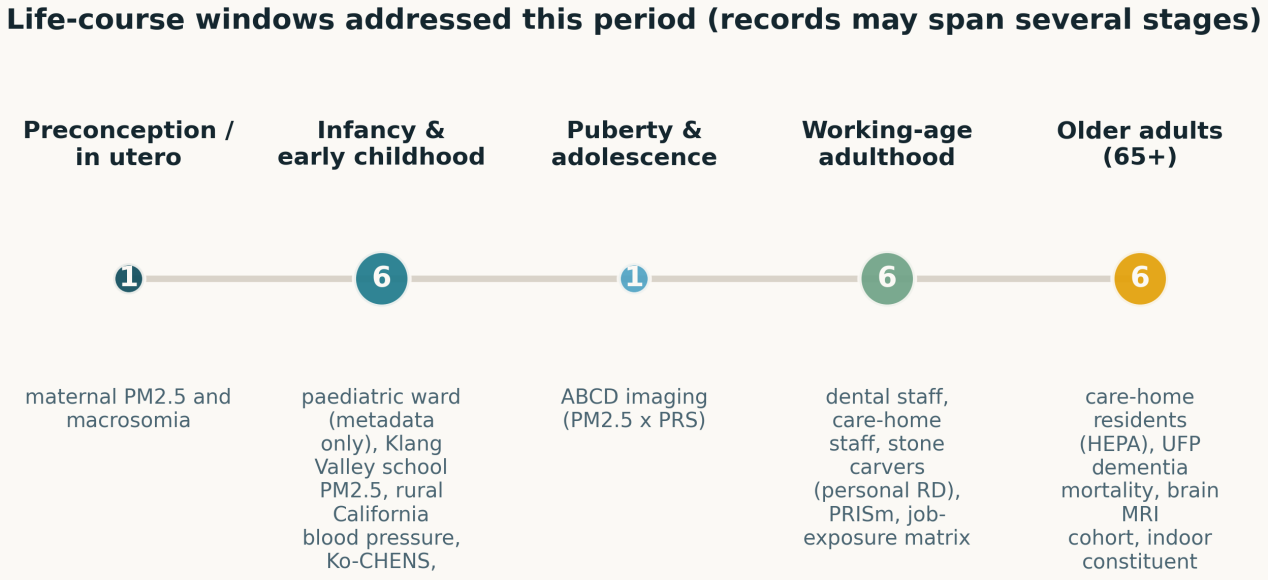


Fig. Life-course windows across the pooled week; counts non-exclusive. **Infancy & early childhood, working age and older adults tie at 6**, with **preconception / in utero at 1** — a sharp reversal from last week, when that window led the series for the first time at 11, and partly an artefact of the held periconception record. **Adolescence 1** (the ABCD imaging analysis) is the weakest window for the seventh consecutive week.



Fig. All **16** pooled estimates, ratio scale, every one with a 95 % CI. **Not mutually comparable and not meta-analysed**: increments include per-IQR, per-5, per-10 and per-1 $\mu g\,m^{-3}$, per 10,000 cm^{-3} and per-lag-window, on HR, OR and RR scales, so **no I^2 is reported** and none would be meaningful. Readable points: the **largest estimates** are the UFP dementia-mortality HR **1.22** (1.17–1.27) per 10,000 cm^{-3} and the chronic-disease incidence HR **1.22** (1.19–1.26) per 5 $\mu g\,m^{-3}$ — different exposures, different diseases, coincidentally equal point estimates, and the first of the two is the only number-concentration estimate in the series. **Two nested families dominate the panel**: four indoor-constituent mortality HRs from one cohort of 12,607 older adults, where black carbon reads **1.072** (1.035–1.111) per $\mu g\,m^{-3}$ against $PM_{2.5}$ mass **1.003** (1.001–1.005), and four source-specific dementia HRs from a second cohort on per-source IQRs. Within each family the estimates share cases and exposure surface and are **nested, not independent**. **One interval crosses unity**: incident CKD per 10 $\mu g\,m^{-3}$ $PM_{2.5}$, 1.22 (0.99–1.50), in the same study whose PM_{10} estimate does not — and whose ozone estimate is inverse, which is the tell that single-pollutant models in that city are not separating anything.

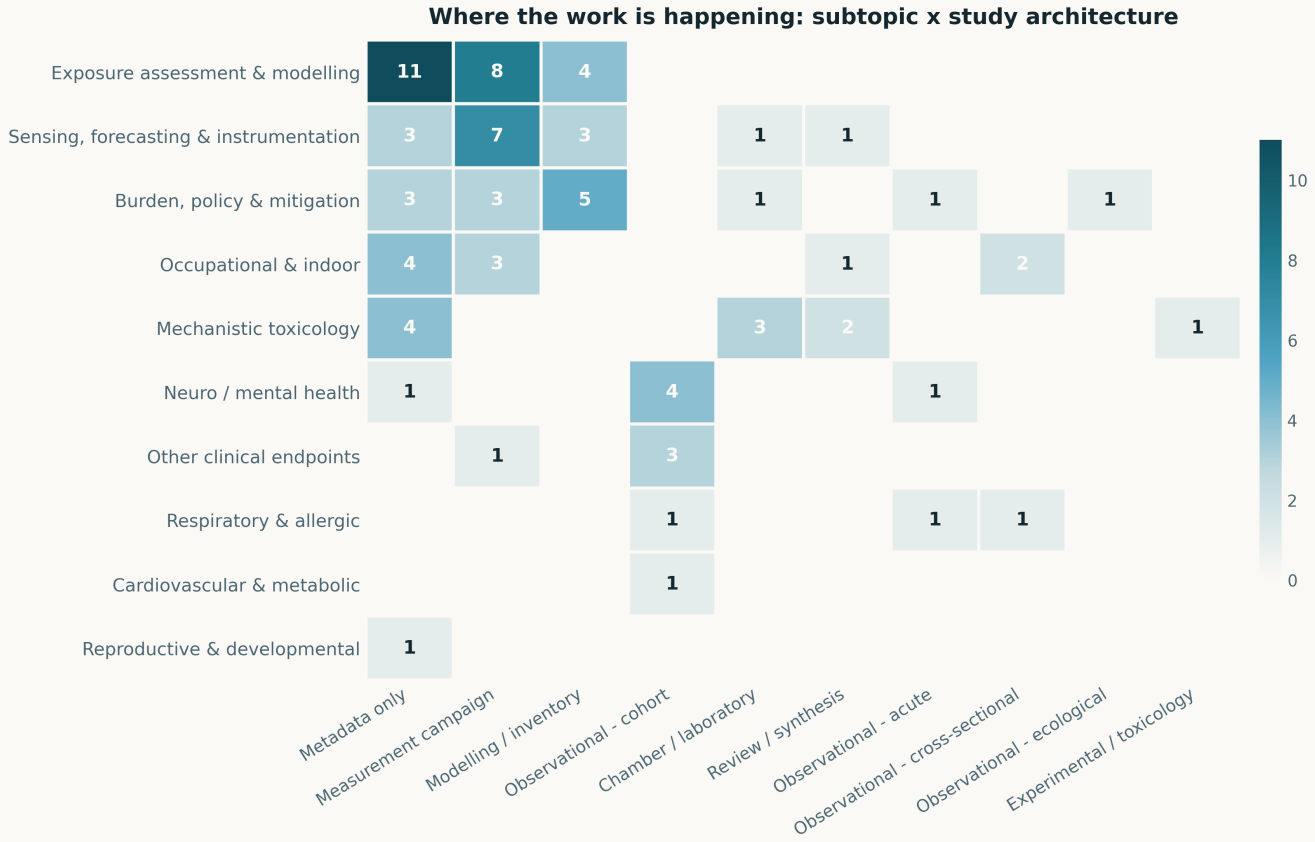


Fig. Subtopic × architecture, pooled. The densest cells are *Exposure* × *Metadata only* and *Exposure* × *Measurement campaign* — the abstract gap stays concentrated on the measurement side, where **14 of 38 records (37 %)** carry no abstract, up from 33 %. The standing finding survives an eleventh week: of the **38** measurement-side records, **one** carries any outcome label and **none observes a human outcome**.

Venues, trial watch, and what the pipeline did

Venues. *Environ Pollut* leads at **7**, then *Atmos Environ* 6, *Atmos Chem Phys* 5, and *Atmosphere*, *Atmos Pollut Res* and *Build Environ* at 4. The leading venue holds 8 %. *Atmos Environ*, *Environ Pollut* and *Build Environ* supplied most of the metadata-only records — Elsevier same-day deposits, for the fourth rollup running. **Preprints supplied 8** (5 from Research Square), down from 11; no medRxiv record entered this week, against seven last week.

Recurring authors. Unchanged: surname frequency is a naming artefact and **no repeat-group cluster is claimed**. No dataset repeated three times; the largest cohorts of the week (2.1 million adults for UFP dementia mortality, 6.1 million Medicare ADRD admissions) each supplied one record.

Trial watch — one new registration. The ClinicalTrials.gov leg was queried for the period on the 12 October run (`LastUpdatePostDate` 4–12 October) and returned **four hits, one kept**: **NCT07500948** (EPIC-AIR, University of Leicester, n=80, interventional, not-yet-recruiting), a stratified feasibility RCT embedding **real-time air-quality guidance** into cardiac and pulmonary rehabilitation. Its `analyze_endpoints` shows three Holm-adjusted co-primaries — weekly time-weighted **modelled ambient** $PM_{2.5}$, PM_{10} and NO_2 during GPS-recorded moderate-to-vigorous activity — with weekly accelerometer-derived activity as the only registered secondary. **The exposure endpoint is a modelled concentration at a GPS location, not a measured personal dose**, which is precisely the substitution the 10 October calibration review warns about; a trial testing whether exposure guidance works should ideally measure what it moved. **NCT07540000** (INTERLUNG) posted a start-date change (15 September → 15 November); **NCT07713264** (VISIONS) was already held; **NCT07850648** (CAN-EXACT) was screened out again for having no PM measurement. `trials.json` now holds **17 trials** over **28 logged windows**, all counters re-derived.

What the harvesting legs did. Crossref-by-ISSN was again the highest-yield leg (6–45 records/day) and again the source of most missing abstracts. **Europe PMC** ran 0–33/day and returned **zero** on 11 October, its first empty day since 25 September. The harvester's **PubMed sensing query returned 0–6 daily**, and the **PubMed connector stayed load-bearing**, appending 1–13 records a day the harvester queries missed. **OpenAlex returned HTTP 429 on every run of the period** — nothing contributed for four weeks; it is now treated as **dead, not degraded**, and is reported as such rather than retried. **arXiv** answered (148 kB) and contributed **one** in-window entry, on 9 October. **Consensus** returned 10 calibration hits and **0 new** every day, for the seventh consecutive week — note that several of its standing hits are cited by the calibration review admitted on 10 October, which is as close as that leg has come to adding value. **Scholar Gateway failed on every attempt** with `ACCESS_DENIED`.

Open gaps

- **The abstract gap is now the single largest defect in the watch. 31 % of the week** shipped without an abstract, the worst reading of the series, and **14 of 27** are measurement-side records — this week including a traceable fast-scan MPSS calibration, an FT-IR speciation study from Pretoria, an adjoint correction for nodal indoor air-quality models and a rural PAH exposure assessment, all of them records this watch exists

to read. **Scholar Gateway has been down for two weeks** and needs reconnecting; the deferred-retry pass remains unbuilt for a fifth week.

- **Mass is the wrong metric, and the week said so three ways.** UFP number, source-resolved deposited dose and nitrogen-containing organic composition all produced results that a $PM_{2.5}$ mass network cannot reproduce. No record this week validated a low-cost channel for any of the three.
- **Calibration does not transfer, with numbers.** Median field R^2 0.72 against laboratory 0.92, median field RMSE **12.3** $\mu g\,m^{-3}$ — and only 9 of 66 reviewed studies reported RMSE at all. Any network claim in this corpus should be read against that 12.3.
- **Uncertainty quantification failed its own preregistered test** under temporal hold-out (coverage 0.794 against a declared [0.85, 0.95] band). Every other exposure surface in the week ships point estimates with no interval at all.
- **One record held, not rejected.** `10.1038/s41467-026-77623-9` (congenital heart defects via circulating proteins), `10.1021/ehp.6c00004` (green space and biological aging) and `10.3389/ftox.2026.1943932` (alveolar organoids) were admitted to the 9 October issue but not authored when this run's authoring was interrupted; all three are logged as held and are **absent from this pool**. `10.1021/acs.est.6c07515`, held from the 3 October issue, is **still unplaced** after one week.
- **None of 38 measurement-side records observes a human outcome**, eleventh consecutive week.
- **Pipeline defects carried forward**: no first-author affiliation field; the `Cancer/Oncologic` split in `ENDPOINT_CANON`; no `\DIGESTDATE`-vs-`--date` build guard; the 2026-07-27 corpus missing `LIFECOURSE`; `.git` lock debris accumulating on the delete-restricted mount. `metrics.csv` continues clean (`csv.writer`, 581 rows at period close, 0 malformed).

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation — 15 records

Carried from the daily issues — full summaries (15 of 15 records in this subtopic)

Calabar ZigBee WSN 2026 Research Square (preprint)

Design and Performance Evaluation of a ZigBee-Based Wireless Sensor Network for CO and $PM_{2.5}$ Air Quality Monitoring in Calabar Metropolis

A ZigBee mesh of low-cost CO and $PM_{2.5}$ nodes, with analytical models for sensing, communication, energy use and dispersion exercised in MATLAB. Reported: reliable links, coverage growing with node count, a lifetime of about 4,065 transmissions, mean CO 30 ppm, mean $PM_{2.5}$ 40 $\mu g\,m^{-3}$, AQI 112.5, and a composite 'performance index' of 76.5%. Limitation: the central weakness is that no calibration, co-location or reference comparison is reported at all - a 40 $\mu g\,m^{-3}$ mean from uncalibrated optical nodes in a humid tropical city is not a measurement, and a self-defined performance index is not a validation; the dispersion component is simulated, not observed. Relevance: moderate as a record of the deployment gap in West Africa, where ambient data is genuinely absent; low as evidence. This is the failure mode this watch exists to flag.

doi: 10.21203/rs.3.rs-10637757/v1

Chang et al. 2026 Environ Pollut

Wintertime vertical stratification of air pollutants and meteorological associations within an urban forest canopy in Shenyang, China

A three-level monitoring mast inside an urban forest in a cold-climate northern Chinese city through the winter heating season. $PM_{2.5}$ and PM_{10} were generally higher at 10 and 20 m than at pedestrian level, O_3 and CO higher at 1.5 m, and NO_2 and SO_2 peaked mid-canopy; relative humidity and wind speed dominated the statistical and machine-learning associations, but height-dependently. Limitation: one site, one winter, no replication mast and no flux measurement, so the inversion of the usual near-ground maximum is described rather than explained - and at wintertime humidity, uncorrected optical PM readings are themselves height-correlated through the humidity profile. Relevance: high - siting guidance for low-cost networks assumes a well-mixed layer below canopy height; this is direct evidence that the 1.5 m reading is not the 20 m reading.

doi: 10.1016/j.envpol.2026.129270

Chiappini WRF-Chem nudging 2026 Atmosphere

Regime-Dependent Effectiveness of Spectral Nudging in WRF-Chem: Meteorological and Saharan Dust Skill over the Central Mediterranean

A two-tier design separates the meteorological response to spectral nudging (a 104-day chemistry-free pair) from the dust-transport response (paired event simulations). Nudging improves sea-level pressure most (daily RMSE 2.08 to 1.53 hPa) and 10 m wind less (2.50 to 2.20 m/s), while 2 m temperature does not improve (2.05 to 2.15 C); the benefit recurs episodically rather than growing with simulation age. Dust skill is conditional: in the weakly forced June case all seven AERONET sites improved (station-mean RMSE 0.457 to 0.235, correlation 0.778 to 0.881), while in March RMSE fell at all eight sites but mean correlation was unchanged. Limitation: two outbreaks is a thin basis for a regime claim, and AOD is a column integral that constrains surface dust PM only weakly. Relevance: moderate - dust is the event class where satellite-and-model PM products diverge most from surface nodes.

doi: 10.3390/atmos17100992

Dai et al. 2026 Atmos Chem Phys

Cross-phase partitioning of sulfur-nitrogen ratios and aerosol mixing-state evolution based on single-particle observations

SPA-MS through winter and summer emission-control periods and matched normal periods, with three sulfur-to-nitrogen metrics (gas gSNR, particle pSNR, number-based nSNR) analysed by causal inference plus interpretable machine learning. Control periods cut NOx while SO_2 stayed roughly flat, raising gSNR, lifting pSNR by about 57% across particle classes and expanding mean vacuum aerodynamic diameter by 27%; enrichment was greatest in particles containing both black and organic carbon and least in BC-free inorganic particles. Limitation: emission-control periods are confounded with season and meteorology, and 'causal inference' on an observational time series with co-varying controls is a strong label for what remains an association. Relevance: high - a 27% diameter shift under a control policy is a direct change in the optical cross-section a nephelometric sensor measures, i.e. a calibration drift caused by policy rather than by hardware.

doi: 10.5194/acp-26-14015-2026

Dayare Navi Mumbai 2026 Preprints.org (preprint)

Weather, Co-Pollutants and Daily $PM_{2.5}$ in a Coastal Indian City: Multi-Station Evidence for Meteorology-Triggered Air-Quality Warnings in Navi Mumbai

VAR models with exogenous weather on screened, unimputed station data. Lagged co-pollutants add little: equivalence tests bound six of seven effects within +/-5% per SD; lagged ozone raises $PM_{2.5}$ by 4%. Rain and boundary-layer height lower same-day $PM_{2.5}$ by 10-24% per SD; date effects shared across stations explain 78% of variance, three-quarters of it weather. A stagnation index plus persistence warns as well as XGBoost, which does not beat persistence. Limitation: a preprint; ERA5 boundary-layer height over a coastal city is coarse, and five stations limit the spatial claim. Relevance: high - a clean demonstration that 'driver' inference from co-movement is weather confounding, and that persistence is the baseline any network forecast must beat.

doi: 10.20944/preprints202610.0209.v1

Fang LCS calibration review 2026 Curr Environ Health Rep

Calibration of Low-Cost Particulate Matter Sensors for Exposure Assessment: A Critical Review

Sixty-six studies, 32 laboratory evaluations and 34 field studies. Median calibrated R2 was 0.92 (n = 31) in the laboratory against 0.72 (n = 29) in the field, and median RMSE 2.6 $\mu g\,m^{-3}$ (n = 7) against 12.3 $\mu g\,m^{-3}$ (n = 9) - a factor of nearly five in error on crossing from bench to deployment. The authors are explicit that these are summaries of a heterogeneous literature, not a pooled effect: aerosol mixture, comparison instrument, concentration range, averaging interval and validation design all differ. RH and temperature were handled in most studies; ML and hybrid models appear more often in field work and sometimes lower error, but independent validation across time, place and sensor unit is uneven. Limitation: only 7 and 9 studies reported RMSE at all, so the headline error figures rest on a small and probably favourable subset. Relevance: high - this is the reference statement that calibration does not transfer, which is the standing assumption behind every distributed network claim in this corpus.

doi: 10.1007/s40572-026-00566-2

Ghajari fuzzy DL Tehran 2026 Research Square (preprint)

A Novel Fuzzy Hybrid Deep Learning Architecture for Spatiotemporal $PM_{2.5}$ Prediction

A fuzzy transformer-CNN-GRU with cross-branch attention for one-day-ahead $PM_{2.5}$ in Tehran, trained on 2014-2015 and tested on 2016, outperforming seven comparator models with a reported RMSE of 0.73 $\mu g\,m^{-3}$. Limitation: an RMSE below 1 $\mu g\,m^{-3}$ for next-day $PM_{2.5}$ in Tehran is implausible for a genuine forecast and suggests leakage (e.g. same-day inputs or normalised units); no persistence baseline is reported, and the data end in 2016. Relevance: low - the same-day Navi Mumbai record shows why persistence must be the reference; this one does not test against it.

doi: 10.21203/rs.3.rs-11058151/v1

Hazelwood et al. 2026 J Aerosol Sci

Size-Resolved Particle Dry Deposition Velocities in a Controlled Spherical Chamber: Effects of Turbulence, Surface Complexity, and Particle Physicochemical Properties

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 5 Oct with no abstract. By title, a chamber determination of size-resolved dry deposition velocity against turbulence intensity, surface complexity and particle physicochemical properties. Deposition velocity is the term that sets indoor particle loss rates and therefore the inferred air-change rate in every filtration study that does not measure one, and it is the largest uncertainty in indoor-outdoor infiltration models. Relevance: potentially high; on the retry list.

doi: 10.1016/j.jaerosci.2026.106903

Loh et al. 2026 Mar Pollut Bull

Seasonal characteristics of submicron aerosols in the coastal atmosphere of southeastern Korea: Influence of urban-port outflow

Seasonal HR-ToF-AMS measurement at a coastal receptor site southwest of Busan, winter 2021 to autumn 2022. PM_1 ranged 4.23-10.0 $\mu g\,m^{-3}$; PMF resolved five organic factors including hydrocarbon-like, oxidised primary (OPOA), two oxygenated and biomass-burning. Ultrafine growth events were frequent, most in autumn, when OPOA reached 36% alongside elevated black carbon under north-easterly transport from the port. Limitation: one receptor site with no upwind reference, so 'port influence' rests on wind sector rather than on a source-resolved tracer, and AMS does not measure refractory sea salt or dust - the submicron mass is non-refractory by construction. Relevance: high - growth events at 4-10 $\mu g\,m^{-3}$ of mass are invisible to a mass-reporting network, which is the recurring theme of this week's issues.

doi: 10.1016/j.marpolbul.2026.120437

Malik MPSS calibration 2026 Aerosol Sci Technol

Traceable calibration of MPSS in fast-scan mode: Recommendations for PNSD measurement at urban traffic hot spots

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 9 Oct with no abstract. By title, a metrology-grade calibration route for MPSS/SMPS operated in fast-scan mode, with recommendations for particle number size distribution at traffic hot spots. Fast scans trade sizing accuracy for time resolution - the scan-direction smearing and DMA transfer-function errors that dominate when concentrations change faster than a scan are the reason roadside PNSD is rarely comparable across sites. Relevance: high - a traceable reference for sub-100 nm number is the missing anchor for any UFP sensor co-location; on the abstract-retry list.

doi: 10.1080/02786826.2026.2726892

Mei et al. 2026 Atmos Chem Phys

Synthesis of the tethered balloon system and other TRACER campaign measurements elucidates aerosol property profiles

149 tethered-balloon flights over greater Houston in summer during the DOE ARM TRACER campaign, with back-trajectory k-means clustering into marine, mixed and urban/long-range air masses. Profiles vary strongly with cluster, modulated by boundary-layer depth and coastal circulation; the marine cluster had CCN at 0.8% supersaturation below 1,000 cm^{-3}, urban and mixed clusters higher. Limitation: one coastal city, one summer, and CCN is inferred from size distribution under an assumed kappa rather than measured, so composition heterogeneity propagates straight into the activated fraction; tethered-balloon sampling is intermittent and fair-weather-biased. Relevance: high - this is the measurement that exposes the central fiction of a surface network, namely that a 2 m reading represents the column a satellite retrieval sees.

doi: 10.5194/acp-26-13885-2026

Metris and Metris 2026 PeerJ

Fungus above us: exploring the atmospheric mycobiome by aircraft

Aircraft bioaerosol sampling across seasons, sequenced and joined to vegetation-productivity and meteorological data. Diversity fell with altitude and with increasing vegetation productivity; decomposers and pathogens dominated, with over 40% of taxa putative plant or animal pathogens (Neoascochyta, Cladosporium, Alternaria, Penicillium among the abundant genera); guild composition shifted with productivity and surface wind. Limitation: a small number of flight tracks stand in for a continent, relative abundance from amplicon data is not concentration, and viability is untested - 'pathogen' here is taxonomic assignment, not infectivity. Relevance: moderate - aircraft campaigns set the vertical context that ground bioaerosol sensors, which cannot resolve taxa at all, are implicitly assuming.

doi: 10.7717/peerj.21746

Pavlidis cold start 2026 Environ Sci: Atmos

Measurements of Real-World Cold Start Emissions in a Parking Garage

Only the deposited lead sentences are available: motor-vehicle cold starts contribute disproportionately to total vehicle emissions, and this work quantifies them under real-world conditions by measuring in a parking garage rather than on a dynamometer. A semi-enclosed garage gives the dilution control a roadside site lacks while keeping real engines, real soak times and real driver behaviour. Limitation: the abstract is truncated at deposit, so emission factors and fleet composition cannot be read; garage ventilation sets the dilution and is rarely characterised well enough to convert concentrations to per-vehicle factors. Relevance: high - cold starts are short, intense, local events, the exact signal a dense node network resolves and a regulatory monitor averages away. The 8 October aged-parking-garage-aerosol record sampled the same kind of microenvironment for oxidative potential.

doi: 10.1039/d6ea00044d

Pires quantum NN 2026 arXiv (preprint)

An Annealing-based Quantum Neural Network with Discrete Wavelet Transforms for Enhanced Air Pollution Nowcasting

Hybrid variational quantum circuits with quantum annealing, scaled to 172 qubits on a neuromorphic platform, nowcast $PM_{2.5}$ from UK urban data and are reported to outperform MLP and LSTM on raw high-entropy inputs, more so with wavelet decomposition. Limitation: the abstract gives no error metric, horizon or persistence baseline, and beating an MLP/LSTM on unpreprocessed data is a weak comparator; 'quantum utility' claims need a classical model tuned on the same wavelet features. Relevance: low - nowcasting at node level is a solved problem for persistence-plus-meteorology.

doi: 10.48550/arXiv.2610.13882

Zhang et al. 2026e Appl Opt

Full field of view temporal calibration for the directional polarimetric camera onboard the GaoFen-5 (02) satellite based on a data reconstruction method

METADATA ONLY. Metadata-only record: PubMed indexed title, authors and DOI on 6 Oct; Applied Optics deposited no abstract and the retry leg recovered none. By title, a full-field-of-view temporal calibration for the DPC instrument, correcting on-orbit response drift by data reconstruction. Polarimetric retrievals are the satellite side of aerosol loading, and a drifting on-orbit calibration propagates straight into the AOD that surface $PM_{2.5}$ models are trained against - the same class of problem as field drift in a ground sensor, at a different scale. Relevance: moderate-high; on the retry list.

doi: 10.1364/ao.597069

Exposure assessment & modelling — 23 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (23 OF 23 RECORDS IN THIS SUBTOPIC)

Aerosol Fe isotopes 2026 Atmos Chem Phys

Isotopic composition of aerosol iron from anthropogenic sources: implications for source apportionment of aerosol iron

Iron isotope endmembers measured for desert dust and several anthropogenic sources: dust averaged +0.14 +/- 0.10 per mil (n=7, consistent with prior work), power-plant coal fly ash +0.26 +/- 0.18 (n=28) and steelwork fly ash -0.07 +/- 0.41, with a wide anthropogenic range overall. Limitation: the endmembers overlap heavily - a steelwork standard deviation of 0.41 per mil against a dust mean of +0.14 means a mixed urban sample cannot be apportioned by delta-56Fe alone, which the spread itself makes plain; small n for dust. Relevance: moderate - soluble aerosol iron matters for oxidative potential and for ocean biogeochemistry, and isotopic apportionment is the kind of offline chemistry that fixed networks will never do but that calibrates what they see.

doi: 10.5194/acp-26-14073-2026

Anwar Pretoria FTIR 2026 Aerosol Air Qual Res

$PM_{2.5}$ Composition and Organic Matter Sources in Pretoria, South Africa Employing Fourier Transform-Infrared Spectroscopy

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 10 Oct with no abstract. By title, FT-IR functional-group analysis of $PM_{2.5}$ organic matter in Pretoria with source attribution. FT-IR on routine filters is one of the few speciation routes cheap enough to run network-wide, and sub-Saharan African composition data are scarce enough that any new site matters. Relevance: high for the composition gap this corpus repeatedly hits; on the abstract-retry list.

doi: 10.1007/s44408-026-00172-9

Arora rural PAH 2026 Atmos Environ

Seasonal PAH Exposure and Health Risks at Central European Rural Background Sites Driven by Combustion Sources and Transport Pathways

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 9 Oct with no abstract. By title, seasonal particle-bound PAH exposure and incremental lifetime cancer risk at Central European rural background sites, attributed to combustion sources and transport. Rural background PAH is dominated by residential solid-fuel burning, the same source the same-day Belgrade apportionment puts at 21% of PM_{10}. Limitation unverifiable from metadata. Relevance: moderate - PAH is a composition endpoint no optical node reaches; on the abstract-retry list.

doi: 10.1016/j.atmosenv.2026.122419

Balabat et al. 2026 World J Microbiol Biotechnol

Urban Road Dust as a Microbial Hotspot: Spatial Variation of Bacterial Communities and Metal Associations in the Philippines

Road dust from nine sites across three metropolitan areas profiled by 16S rRNA metabarcoding with PICRUSt2 functional prediction and trace-metal pollution indices. Proteobacteria and Actinobacteria dominated everywhere with a persistent core of Paracoccus, Acinetobacter, Kocuria and Deinococcus; alpha diversity did not differ between cities (Kruskal-Wallis p>0.05) but composition did (PERMANOVA R2=0.466, p=0.004). Paracoccus rose toward the most metal-laden Cagayan de Oro site and correlated with cobalt and nickel alongside raised predicted denitrification. Limitation: three sites per city, no replication within site, PICRUSt2 infers function rather than measuring it, and road dust is a reservoir - nothing here shows what becomes airborne or is inhaled. Relevance: low-moderate - the resuspended coarse fraction that low-cost optical nodes do see is biologically non-inert.

doi: 10.1007/s11274-026-05307-z

Budapest deposition 2026 Environ Sci Technol

Aerosol Deposition in the Human Respiratory System from Distinct Urban Sources Using Multiple Exposure Metrics

Source apportionment of particle number at an urban background site feeds a lung deposition model under reference breathing conditions. Most particles reach the deep lung with maximum deposition at airway generations 17-21 depending on activity; deposition fraction varies by source from 76% (new particle formation) to 27% (solid fuel combustion). Exercise raises total deposition and shifts it from extra-thoracic to alveolar; deposition rates are more sensitive to activity than deposition fractions, and number and surface-area curves are similar in shape but rank sources differently. Limitation: a deposition model, not a measurement - ICRP-type models are calibrated on healthy adults at rest and carry large uncertainty below 20 nm, exactly where the new-particle-formation fraction sits. Relevance: high - a factor of nearly three in deposited fraction between sources at the same ambient number concentration is the clearest statement in this backfill that mass, and even number, is not dose.

doi: 10.1021/acs.est.6c10850

Carnec adjoint nodal IAQ 2026 Indoor Environments

Adjoint-based corrective approach for nodal indoor air quality predictions

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 9 Oct with no abstract. By title, an adjoint-based correction for nodal indoor air quality models, which treat each room as one well-mixed node and are fast but systematically wrong where a room is not well mixed. An adjoint formulation gives sensitivity of predicted concentrations to each input at the cost of one extra solve, which is what makes correction from sparse sensor data tractable. Relevance: high - this is the data-assimilation route by which a handful of indoor nodes can correct a whole-building model, the indoor analogue of the conformal-coverage question raised on 8 October. On the abstract-retry list.

doi: 10.1016/j.indenv.2026.100199

Chaudhary India meteorology 2026 Environ Monit Assess

Meteorological modulation of ground-level air quality in India based on in situ measurements

Eight years of in-situ measurements across Indian monitoring sites with Sen's slope trends. $PM_{2.5}$ fell at several Indo-Gangetic sites (-0.893 to -0.097 $\mu g\,m^{-3}$/month) and PM_{10} likewise (-1.320 to -0.169), while Punjab rose on local sources; ozone increased (0.041 to 0.529 $\mu g\,m^{-3}$/month), read as aerosol inhibition weakening plus more precursors. Declining boundary-layer height and wind speed over north and east India modulate all of it. Limitation: Sen's slope on eight years of CPCB data cannot separate emission change from meteorological trend, which is the paper's own point turned against its trend estimates; station siting and instrument changes over that period are undocumented. Relevance: moderate - the regional version of the same-day argument that a concentration trend is not an emissions trend.

doi: 10.1007/s10661-026-15915-9

Chavan and Lataye 2026 Environ Sci Pollut Res Int

Application of Openair package in R programming and PCA: unveiling air quality dynamics and meteorological influences

A full year of regulatory monitoring data for a tier-2 Indian city analysed with openair and PCA. Winter AQI reached 492 (severe) in February with monsoon improvement; varimax PCA gave two components explaining 73.1% of variance, read as anthropogenic combustion (PM, NO_2) and photochemistry (O_3, temperature); $PM_{2.5}$ and PM_{10} exceeded CPCB standards on 27-29 days per winter month. Limitation: descriptive reanalysis of one station-year with no source apportionment, no receptor model and no validation - PCA components here are correlation structure, not sources, and the policy recommendations do not follow from the analysis. Relevance: low as method, moderate as a reminder that tier-2 cities have one regulatory station and no spatial information at all.

doi: 10.1007/s11356-026-38275-w

Conformal $PM_{2.5}$ mapping 2026 Research Square (preprint)

Preregistered External Validation of Spatially Valid Conformal Prediction for Air-Pollution Exposure Mapping in China

A preregistered external validation - rare in this literature - of spatially valid conformal prediction against gradient boosting, with an acceptance band of $[0.85, 0.95]$ on pooled coverage declared in advance. Under spatial hold-out the spatially weighted method reached 0.873 coverage at nominal 90% (gradient boosting: 0.486), 5 of 7 macro-regions met the block criterion (worst, North China, 0.719), and interval widths matched equal-weight conformal prediction, so coverage was not bought with wider intervals. Under temporal hold-out coverage fell to 0.794, outside the band, and the preregistered joint hypothesis failed - traced to the sharp post-2017 $PM_{2.5}$ decline. Limitation: province-level annual means are a very coarse spatial unit; coverage there says little about a 1 km surface, and the temporal failure is the one that matters for forecasting. Relevance: high - every exposure map in this corpus ships point estimates with no interval, and this is the first record in the series to preregister the test and report the failure.

doi: 10.21203/rs.3.rs-11182286/v1

Fazelvalipour LA microplastics 2026 Preprints.org (preprint)

Size-Resolved Characteristics and Source Influences of Airborne Microplastics in Central Los Angeles

Two seasonal campaigns in central Los Angeles: microplastics were in every sample at 2.2-4.0 MP/m3 (cool season) and 2.2-3.4 MP/m3 (warm), dominated by fibres and by the 5-50 and 50-100 um size fractions. Total MP correlated strongly (rho >= 0.79) with crustal (Fe, Si, Mn, Ca, Al, Mg), traffic (Ba, Cu, Ni) and carbonaceous (OC, EC) tracers, while short-term meteorology showed no consistent relationship, implying MPs travel as part of the resuspended urban particle mixture rather than as a separate contaminant. Limitation: a preprint; microscopy-based MP counts carry large operator and blank uncertainty, correlation with crustal tracers is also what resuspension artefact looks like, and particles under 5 um - the inhalable fraction - are not resolved. Relevance: moderate-high - supports the same-day tyre-wear argument that the growing non-exhaust, coarse, resuspended fraction is what networks increasingly measure badly.

doi: 10.20944/preprints202610.0577.v1

GOES-R $PM_{2.5}$ CONUS 2026 GeoHealth

Machine Learning Predictions of $PM_{2.5}$ and Their Applications to Exposure and Health Assessment in the CONUS

A two-phase framework: a U-Net-like partial convolutional network inpaints missing geostationary AOD (agreement with AERONET R=0.75, paired observations raised from 430,603 to 728,115 by iterative gap-filling), then a random forest predicts hourly surface $PM_{2.5}$ at ~1 km for 2019 onward. Limitation: the inpainting is validated against AERONET, which is sited where retrievals already work - gap-filled cells are by definition cloudy, humid or smoke-covered, and their skill is not demonstrated; a random forest on gap-filled inputs inherits that uncertainty without propagating it. Relevance: high - hourly rather than daily is the step that makes a satellite product comparable to a sensor network's native time resolution, and the same day's conformal-prediction record is the missing half of this one.

doi: 10.1029/2026gh001884

Hayli Gubbi eruption 2026 Atmos Environ

Short-term Atmospheric Impacts of the Hayli Gubbi Volcanic Eruption 2025

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 7 Oct with no abstract. By title, the short-term atmospheric impact of the 2025 Hayli Gubbi eruption. Volcanic SO_2 and ash plumes are the natural experiment that tests whether a sparse monitoring network can detect and attribute a known, dated, large perturbation - and East African ground monitoring is close to absent, so the answer is informative either way. Relevance: moderate; on the retry list.

doi: 10.1016/j.atmosenv.2026.122406

Hopke et al. 2026 Atmos Pollut Res

Characterization of PM Pollution Episodes in Kuwait

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 5 Oct with no abstract. By title, a characterisation of PM pollution episodes in Kuwait. Dust-dominated Gulf environments are where optical low-cost sensors fail hardest - coarse mineral dust sits outside the calibrated size and refractive-index range of nephelometric nodes - and the episode statistics are the input any regional network design needs. Relevance: potentially high; on the retry list.

doi: 10.1016/j.apr.2026.103226

Mukhamediyarov et al. 2026 Atmosphere

Elemental Composition, Sources, and Health Risks of Ambient $PM_{2.5}$ in Ust-Kamenogorsk (Oskemen) and Pavlodar, Kazakhstan

PCA separates a non-ferrous metallurgical profile in Ust-Kamenogorsk from coal fly ash, ferrous metallurgy and petrochemicals in Pavlodar. The risk arithmetic then reports cumulative hazard indices exceeding 1 by more than 500-fold, driven by Zn (HQinh 304.7) and Al (129.5), and carcinogenic risks for Cr and As up to 9x the 1e-4 threshold. Limitation: a hazard quotient of 300 for zinc is a sign that the reference concentration is being misapplied, not that residents are inhaling a toxic zinc dose - EPA inhalation RfCs do not exist for several of these elements and the substitutes used are oral values scaled by default breathing rates; sampling duration and n are not stated. Relevance: moderate for the Central Asian composition data, which is genuinely scarce; the risk numbers should not be propagated.

doi: 10.3390/atmos17100977

N-containing OA 2026 Atmos Environ

Unveiling the Precursor-Specific Aqueous Formation Mechanisms of Nitrogen-Containing Organic Aerosols

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 7 Oct with no abstract. By title, a mechanistic account of how particular precursors form nitrogen-containing organic aerosol in the aqueous phase. Aqueous-phase secondary formation is humidity-dependent, which makes it the physical process underneath the humidity correction every optical low-cost sensor applies - the correction treats growth as hygroscopic swelling when some of it is mass that was not there before. Relevance: moderate-high; on the retry list.

doi: 10.1016/j.atmosenv.2026.122408

$PM_{2.5}$ and GPP 2026 Environ Pollut

Climate-dependent nonlinear relationships between $PM_{2.5}$ and ecosystem productivity across China's urban agglomerations

Satellite observations and interpretable machine learning relate $PM_{2.5}$ to gross primary productivity across climate and pollution gradients. The relationship is robust everywhere but changes sign by region: moderate aerosol loading goes with enhanced productivity in arid environments, while elevated $PM_{2.5}$ generally suppresses it elsewhere, jointly shaped by climate and pollution regime. Limitation: the authors flag the central problem themselves - $PM_{2.5}$, radiation, humidity and temperature co-vary, so a diffuse-radiation fertilisation effect cannot be separated from co-varying meteorology by feature attribution on observational data; both $PM_{2.5}$ and GPP are satellite products with shared retrieval dependencies. Relevance: low for health, moderate as a reminder that aerosol has non-health consequences that enter policy arithmetic.

doi: 10.1016/j.envpol.2026.129305

Petrovic Belgrade PMF 2026 Atmos Chem Phys

Chemical characterization and source apportionment of PM_{10} in Belgrade, Serbia: influence of local and regional anthropogenic and natural sources

USEPA PMF 5.0 on PM_{10} at an urban-background site using 34 inputs including organic tracers - broader than earlier Belgrade apportionments that used elements or ions alone. Biomass burning leads at 21% of PM_{10}, concentrated in the heating season, then ammonium sulphate 18%, mineral dust 17% and mixed traffic/industry 15%, with primary biological and biogenic secondary factors resolved separately. Limitation: one site, and a mixed traffic-industry factor means PMF could not separate the two sources most amenable to local control; PM_{10} mass weighting under-represents the fine combustion fraction that drives health effects. Relevance: moderate - a 21% residential biomass share in a Balkan capital is the case where low-cost optical networks track the winter smoke signal well but need composition-aware RH correction.

doi: 10.5194/acp-26-14165-2026

Rocha Volta Redonda 2026 Environ Monit Assess

Long-term trends and seasonal variability of particulate matter (PM_{10} and $PM_{2.5}$) in Volta Redonda (RJ), Brazil: a 15-year temporal analysis

METADATA ONLY. Metadata-only record: Springer deposited title, authors and DOI on 9 Oct with no abstract, and the retry chain (PubMed, Europe PMC, Semantic Scholar) recovered none. By title, a 15-year PM_{10} and $PM_{2.5}$ trend and seasonality analysis for a steel-industry city in Rio de Janeiro state. Long regulatory records from Latin American industrial cities are scarce, and instrument changes over 15 years are the usual confounder of any trend claim. Relevance: moderate; on the abstract-retry list.

doi: 10.1007/s10661-026-15988-6

Shi et al. 2026 Environ Pollut

Compositional transition and coupled formation pathways of $PM_{2.5}$ pollution in a typical industrial city: An empirical study based on Handan, China

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 5 Oct with no abstract. By title, an empirical account of how $PM_{2.5}$ composition in a heavily industrial Hebei city has shifted and how its formation pathways couple. A compositional transition from primary to secondary is the slow change that silently alters the relationship between an optical sensor's scattering signal and gravimetric mass over years of a network's life. Relevance: moderate; on the retry list.

doi: 10.1016/j.envpol.2026.129296

Wang et al. 2026d Atmos Environ

Physics-inspired inductive spatiotemporal kriging for $PM_{2.5}$ with satellite gradient constraints

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 6 Oct with no abstract. By title, this is the central methodological problem of a sparse network - interpolating to unmonitored locations - attacked inductively (so that new sensors can be added without retraining) and constrained by satellite gradients rather than by a stationarity assumption. The Everglades record in the 2 October issue measured exactly how badly ordinary kriging and IDW fail day to day; this is the proposed repair. Relevance: high, the highest of any metadata-only record in this backfill; flagged for a full-text read as soon as the abstract appears.

doi: 10.1016/j.atmosenv.2026.122398

Wu NCP brown carbon 2026 Atmos Chem Phys

Light absorption properties and composition of water-soluble brown carbon in North China Plain: implication for an enhancing role of nitrogenous organic compounds

Simultaneous winter sampling at Beijing, Tianjin, Luancheng (rural), Handan and Jinan. Abs365 at the rural site was 1.1-3.5 times the urban values, but mass absorption efficiency peaked in Jinan (1.40 m2/g). Urban Abs365 fell ~45% from 2018 to 2023. BrC absorptivity rose from clean to haze with the N:C ratio, pointing to nitrogenous organics as chromophores. Limitation: water-soluble extracts miss the methanol-soluble, more absorbing BrC, and one winter cannot separate trend from meteorology. Relevance: moderate - the same nitrogenous organics the same-day Sci Adv record ranks most toxic are the ones that absorb at 365 nm, which a multi-wavelength absorption node can see.

doi: 10.5194/acp-26-14229-2026

Yadav ENSO India 2026 Atmos Pollut Res

Air Quality in Flux: The Baffling ENSO Transition in India

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 9 Oct with no abstract. By title, an analysis of how an ENSO phase transition altered Indian air quality. Interannual meteorological modes are the main reason a national PM trend over a few years is not an emissions trend - the same-day Navi Mumbai record makes the station-scale version of this point, and the 10 Oct Indo-Gangetic trend record the regional one. Relevance: moderate; on the abstract-retry list.

doi: 10.1016/j.apr.2026.103231

Zhao et al. 2026c Atmos Pollut Res

Spatial Heterogeneity and Multi-scale Statistical Associations of Urban Air Pollution Revealed by Interpretable Machine Learning in Shenzhen, China

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 5 Oct with no abstract. By title, an interpretable machine-learning treatment of intra-urban spatial heterogeneity at several scales in Shenzhen. Scale dependence of the predictor set is the question that decides how dense a network has to be before adding nodes stops buying information, so the result matters more than the method. Relevance: potentially high; on the retry list.

doi: 10.1016/j.apr.2026.103230

Cardiovascular & metabolic — 1 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (1 OF 1 RECORDS IN THIS SUBTOPIC)

Children's AIRE 2026 Int J Hyg Environ Health

Ambient particulate matter exposure and children's blood pressure in rural California, 2019-2022

592 children aged 6-8 at enrolment, blood pressure measured up to five times through spring 2022, linear mixed models. An IQR increase (38 days) in prior-year days above 50 $\mu g\,m^{-3}$ daily PM_{10} was associated with +1.1 mmHg DBP (95% CI 0.8, 1.5) and +0.9 mmHg SBP (0.3, 1.5); the top quartile of short-term $PM_{2.5}$ and PM_{10} carried 5.8 (0.4, 11.2) and 5.9 (0.2, 11.6) percentage-point higher risk of elevated or hypertensive SBP against the lowest quartile. Limitation: exposure comes from regulatory monitors in a rural region where the nearest monitor can be tens of kilometres away and agricultural and wildfire sources are highly local, so non-differential misclassification is large and almost certainly biases toward the null; paediatric BP is noisy across visits. Relevance: high - this is the archetypal study whose exposure term a rural low-cost network would transform, and the 'days above threshold' metric suits an inexpensive, lower-accuracy node better than an annual mean does.

doi: 10.1016/j.ijheh.2026.114923

Neuro / mental health — 6 records

Carried from the daily issues — full summaries (6 of 6 records in this subtopic)

ABCD air pollution 2026 Schizophr Bull

The Relationship between Air Pollution Exposure, Genetic Liability for Schizophrenia, and Brain Outcomes in the ABCD Study

1,580 adolescents with genotype, age 9-10 exposure and structural plus resting-state MRI at 13-14; linear mixed-effects models with a schizophrenia polygenic score as moderator. Higher $PM_{2.5}$ was associated with smaller right frontal pole surface area (p-FDR 0.039); the PRS interactions were non-monotonic - among low-PRS individuals, higher $PM_{2.5}$ went with larger bilateral caudate volumes, while low-PRS/medium-$PM_{2.5}$ showed smaller left caudate than medium/medium. Higher NO_2 was associated with stronger negative connectivity between networks. Limitation: a single surviving FDR-corrected main effect across a very large imaging search space, and interaction patterns that do not order monotonically in either PRS or exposure, is the signature of noise - the authors' own 'first test' framing is the right reading. Relevance: moderate - the exposure is a modelled annual surface at the residence of adolescents who move around a city all day.

doi: 10.1093/schbul/sbag138

Batisse et al. 2026 Epidemiology

Long-term exposure to ultrafine particles and mortality from neurodegenerative diseases: a population-based cohort study

2.1 million adults in the Canadian Census Health and Environment Cohorts, 2001-2019, 19 million person-years and 20,560 neurodegenerative deaths. Per 10,000 particles/cm3 of UFP number, dementia mortality HR 1.22 (95% CI 1.17-1.27), strengthened after adjustment for UFP size; positive for every dementia subtype, largest for vascular dementia. Models were run with and without inverse-probability censoring weights for competing events and adjusted for co-pollutants. Limitation: postal-code exposure from a land-use-regression UFP surface, no mobility or indoor time, and cause of death on the death certificate under-ascribes dementia; UFP number and NO_2 share the traffic gradient, so co-pollutant adjustment cannot fully separate them. Relevance: high - UFP is unregulated and unmeasured by every optical low-cost node, which sees mass, not number; this is the strongest current argument for CPC-class instruments in dense networks.

doi: 10.1097/ede.0000000000002054

Byun ADRD readmission 2026 Environ Health Perspect

$PM_{2.5}$ and 30-Day Readmission in Medicare Beneficiaries with Alzheimer's Disease and Related Dementias: Effect Modification by Neighborhood Green and Blue Space

Nationwide time-stratified case-crossover over 6,098,836 ADRD index admissions, 759,128 (12.4%) readmitted within 30 days. Per IQR (7.2 $\mu g\,m^{-3}$) of lag 0-2 $PM_{2.5}$: OR 1.010 (1.006-1.015), stronger for readmissions after day 7. Landsat- and Street-View-derived green and blue space did not conclusively modify the association. Limitation: ZIP-level modelled $PM_{2.5}$ for a largely homebound or institutionalised population is a poor proxy for the air actually breathed, which biases toward the null and makes the null modification uninterpretable. Relevance: moderate - a 1% per-IQR effect on a 6-million-person denominator is the scale at which indoor sensing in care settings would change the exposure estimate more than the model does.

doi: 10.1021/ehp.6c00147

$PM_{2.5}$ and Mn biomarkers 2026 Environ Pollut

Differential associations of $PM_{2.5}$ and particulate manganese with blood-based amyloid and tau biomarkers

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 7 Oct with no abstract. By title, $PM_{2.5}$ mass and a specific metal constituent are contrasted against blood-based amyloid and tau - the preclinical Alzheimer biomarkers. Separating a constituent from total mass against a mechanistic biomarker is precisely the design the UFP dementia-mortality cohort (5 October) and the nitrate lung-cancer cohort (3 October) both point toward. Relevance: high; on the retry list.

doi: 10.1016/j.envpol.2026.129304

Tang source PM dementia 2026 Eur Arch Psychiatry Clin Neurosci

Associations of long-term exposure to source-specific $PM_{2.5}$, lifestyle, and genetic susceptibility with dementia risk: a prospective cohort study

Cox models with time-varying source-specific $PM_{2.5}$; median follow-up 11.9 years, 5,648 incident dementia cases. Per IQR: domestic HR 1.05 (1.01-1.08), road transport 1.12 (1.04-1.21), other transport 1.12 (1.06-1.19), industry 1.07 (1.03-1.11). High exposure plus high polygenic risk gave a 236-352% excess versus the lowest-lowest stratum. Limitation: source fields come from a dispersion model whose sources are spatially collinear, and per-IQR scaling differs by source, so the ranking of transport over domestic is not a per-ug comparison; the joint-effect excess mostly restates genetic risk; the abstract does not name the cohort or country. Relevance: moderate - source attribution at residence is what a speciated or BC-capable node network could provide directly.

doi: 10.1007/s00406-026-02391-y

Wang et al. 2026c Environ Pollut

Long-Term Air Pollution Exposure and Brain Structural Decline: A Longitudinal Analysis

4,276 middle-aged and older adults with repeated brain MRI; global, grey- and white-matter volumes and white-matter hyperintensity burden regressed on modelled residential $PM_{2.5}$ and PM_{10} in fully adjusted multivariable models. Higher long-term particulate exposure was associated with accelerated global and regional grey-matter loss and greater WMH progression, which the authors read as both a neurodegenerative and a cerebrovascular pathway. Limitation: the abstract gives no effect sizes, no confidence intervals, no exposure contrast and does not name the cohort or country - unusual for a quantitative imaging analysis, and it makes the result unassessable as reported; repeat-MRI subsamples are also selected for survival and compliance. Relevance: moderate - brain structure is the mediator between the UFP dementia-mortality signal and clinical disease, but this record cannot be entered as an effect estimate.

doi: 10.1016/j.envpol.2026.129269

Respiratory & allergic — 3 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (3 OF 3 RECORDS IN THIS SUBTOPIC)

Howlett-Downing and Wichmann 2026 Int J Environ Health Res

Triangulating health risks of $PM_{2.5}$ and trace elements in Pretoria: evidence from health risk assessment, case-crossover, and concentration-response function analyses

Three methods applied to the same secondary dataset: hazard quotients, a case-crossover analysis and concentration-response functions. Hazard quotients exceeded unity in adults, children and infants; the case-crossover OR was 1.027 (95% CI 1.006-1.049) per 10 $\mu g\,m^{-3}$ $PM_{2.5}$. Meeting the South African standard would avert 158 admissions annually, rising to 342 and 748 under the WHO daily and annual guidelines; element-specific functions give 110-608. Limitation: the three methods are not independent evidence - they re-use one exposure series from a small number of sampling days, and summing element-specific attributable cases double-counts the mass they are part of; hazard quotients are screening-level and not comparable to epidemiological risk. Relevance: moderate - Southern African monitoring is thin, and a saturation low-cost network would resolve the within-city contrast this design cannot.

doi: 10.1080/09603123.2026.2733112

Ko-CHENS 36 months 2026 Allergy Asthma Immunol Res

Exposure to Environmental Pollutants and the Risk of Respiratory Illnesses in 36-Month-Old Children: The Korean Children's Environmental Health Study

4,059 children with measured indoor pollutants at a 36-month home visit, blood lead, mercury and cadmium, urinary endocrine-disruptor metabolites and questionnaire outcomes, analysed by quantile g-computation. Allergic rhinitis was most prevalent; children with respiratory illness had more tobacco-smoke exposure and pet ownership, and higher dust-mite levels with recurrent wheezing and pneumonia. The mixture was positively associated with allergic rhinitis (driven by lead and phthalates) and with pneumonia (phthalates). Limitation: the indoor PM measurement is a single home visit standing in for three years of exposure, and in a mixture model dominated by biomarkers with month-long half-lives, that one spot measurement is the weakest term - which is a likely reason PM does not drive any of the mixture associations. Relevance: high - a continuously logging indoor node would replace the single visit, and this cohort shows what the single visit costs.

doi: 10.4168/aair.2026.18.5.668

Sao Paulo paediatric 2026 Front Pediatr

Climate and air pollution are associated with pediatric respiratory hospitalizations in Sao Paulo, Brazil

Paediatric emergency visits and admissions recorded in 2022, geocoded, with exposures assigned by inverse-distance weighting and multivariable logistic models for two outcomes. Higher ozone (OR 1.012, 1.005-1.019), higher relative humidity (1.016, 1.005-1.027) and spring were associated with greater odds that a respiratory encounter was a bronchiolitis admission; lower temperature was associated with the unspecified-ARI outcome (0.887, 0.793-0.993 per unit). Limitation: the outcome is conditional on presenting - these are odds of diagnosis given an encounter, not risk of disease, so the design cannot speak to incidence; IDW over a sparse monitor network in a metropolitan area of 21 million is the same interpolation problem the 2 October Everglades record quantified. Relevance: moderate - no PM estimate reaches significance here, which is itself informative about what IDW exposure does to a signal.

doi: 10.3389/fped.2026.1899313

Reproductive & developmental — 1 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (1 OF 1 RECORDS IN THIS SUBTOPIC)

Qiang maternal PM macrosomia 2026 Chin Med J (Engl)

Interaction of maternal $PM_{2.5}$ exposure and psychosocial state on newborn macrosomia: A nationwide population-based study

METADATA ONLY. Metadata-only record: the publisher deposited title, authors and DOI on 10 Oct with no abstract, and the retry chain recovered none. By title, a nationwide study of the interaction between maternal $PM_{2.5}$ exposure and maternal psychosocial state on newborn macrosomia. Interaction claims between a modelled environmental exposure and a self-reported psychosocial measure are the design most prone to residual confounding by socioeconomic position, and macrosomia is a thresholded birthweight outcome whose prevalence is sensitive to that. Relevance: moderate - birth cohorts are where personal rather than residential exposure measurement would change the estimate most. On the abstract-retry list.

doi: 10.1097/cm9.0000000000004256

Mechanistic toxicology — 10 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (10 OF 10 RECORDS IN THIS SUBTOPIC)

Aged traffic particles 2026 Environ Sci Technol

Aging of Urban Traffic Fine Particles Modulates Toxicity Mechanisms and Oxidative Stress Responses

Real cold-start and urban emissions sampled in a parking garage, aged in an oxidation flow reactor. Ageing raised SOA mass up to 6.9-fold, O:C from 0.26 to 0.79, and both intrinsic and volume-normalised oxidative potential, while peroxide-sensitive DCF activity fell. Aged extracts raised cellular ROS in lung and liver cells and produced greater oxidative-stress and DNA-damage responses; fresh extracts instead suppressed cellular respiration, especially in HepG2, with distinct cell-painting morphological profiles. Limitation: extracts, not whole aerosol, so the soluble fraction is tested and the particle is not; the divergence between acellular DCF and cellular ROS shows how assay-dependent 'oxidative potential' remains. Relevance: high - oxidative potential is the leading candidate for a health-relevant metric a sensor network could one day report, and this is a direct demonstration that it changes downwind of the source.

doi: 10.1021/acs.est.6c06323

Env epigenetics review 2026 Front Epigenet Epigenom

Environmental epigenetic modifiers in neurodegenerative diseases: a systematic review of epigenomic mechanisms and therapeutic targets

A PRISMA-structured review of DNA methylation, histone modification and non-coding RNA changes linking environmental neurotoxicants - particulate matter among them - to neurodegeneration, with emphasis on developmental windows and life-course exposure. Limitation: PRISMA structure without quantitative synthesis, risk-of-bias assessment or any pooling, over a literature in which publication bias toward positive methylation findings is severe; particulate matter is treated interchangeably with metals and solvents despite incomparable exposure metrics. Relevance: low as evidence. Entered because the epigenetic mediator claim keeps appearing beside the dementia-mortality results in this backfill and it should be visible how thin the review base under it is.

doi: 10.3389/freae.2026.1833533

Gao et al. 2026 Carcinogenesis

Air pollution and early-onset lung cancer: from environmental carcinogenesis to early intervention strategies

Narrative review of mechanisms linking air pollution to early-onset lung cancer: genomic instability (DNA damage, telomere dysfunction, mitochondrial impairment, mutational signatures), epigenetic dysregulation, a tumour-promoting niche through CXCL13 and IL-1beta, immune evasion via PD-L1 and CD47-SIRPalpha, and respiratory-gut microbiome disruption, closing with candidate biomarkers and chemopreventive targets. Limitation: no systematic search or evidence grading, the 99%-above-guideline and >14%-of-lung-cancer-deaths framing is quoted rather than re-derived, and the biomarkers are called 'validated' on the strength of single studies. Relevance: low as evidence, moderate as a map of which mediators a cohort with measured personal exposure could actually test.

doi: 10.1093/carcin/bgag070

Melatonin DPM2.5 2026 Ecotoxicol Environ Saf

Melatonin protects keratinocytes from diesel-derived $PM_{2.5}$-induced damage by suppressing JNK/p38 MAPK signaling and caspase activation

Diesel $PM_{2.5}$ raised intracellular ROS, damaged macromolecules, disrupted mitochondrial membrane potential and drove apoptosis in HaCaT cells; melatonin scavenged radicals, restored viability and membrane potential, partly restored Bcl-2 against raised Bax and Bim, and suppressed caspase-9 and -3 and JNK/p38 MAPK activation, with a hairless-mouse arm. Limitation: the same template as the curcumin-nanosphere record on 3 October - a protective-agent paper in which the PM arm is a generic oxidant stimulus, with no composition, no dose-response against a deposited-dose estimate and no endotoxin control; diesel PM is a reference material, not an ambient aerosol. Relevance: low as mechanism; logged to keep the dermal-endpoint strand complete and because the DPM2.5 reference material is at least characterised, unlike most such papers.

doi: 10.1016/j.ecoenv.2026.120879

Mountain West wood smoke 2026 Part Fibre Toxicol

Household wood smoke exposure, lung black carbon deposition, and genomic instability in U.S. Mountain West residents

METADATA ONLY. Metadata-only record: Crossref deposited title, authors and DOI on 7 Oct; neither PubMed, Europe PMC nor Semantic Scholar held an abstract, and Scholar Gateway is still refusing. By title this is an unusually direct chain - a household exposure, a measured internal dose (lung black carbon deposition) and a genomic endpoint in the same human subjects. Internal dose is what every ambient-monitoring epidemiology study substitutes a model for, so a paper that measures it is worth the retry. Relevance: high, the highest of the day's metadata-only records; flagged for full-text read.

doi: 10.1186/s12989-026-00702-8

Sharma et al. 2026 Aerosol Sci Technol

Characterization of carbonaceous particles generated from three flame modes using a single burner to enable properties-based exposure studies

METADATA ONLY. Metadata-only record: Taylor & Francis deposited title, authors and DOI on 6 Oct with no abstract. By title, a single burner operated in three flame modes to generate carbonaceous particles of controlled and differing properties, explicitly to support properties-based rather than mass-based exposure studies. That framing is the open problem behind every toxicology record in this backfill: dose expressed as mass hides composition, and a generator that varies one property at a time is how it gets separated. Relevance: high; on the retry list.

doi: 10.1080/02786826.2026.2737177

Tampere SOA toxicity 2026 Environ Res

From primary to secondary aerosol: chemical transformation and toxicity of gasoline vehicle exhaust under contrasting driving conditions

A Euro 6 plug-in hybrid with a gasoline particulate filter run over a mild RDE cycle and a highly dynamic Combined cycle, cold and hot, with diluted exhaust aged in the Tampere Secondary Aerosol Reactor and delivered to A549 cells through a mobile air-liquid-interface system, fresh and aged, whole exhaust and gas-phase only. Ageing consistently shifted aerosol composition and toxicity relative to primary emissions. Limitation: one vehicle, one fuel, and an OFR delivers days of equivalent ageing in minutes at OH concentrations far above ambient, which over-produces highly oxygenated products; ALI dose at the cell surface is rarely quantified in the same units as an inhaled dose. Relevance: high - a GPF-equipped modern vehicle emitting little primary PM and substantial secondary aerosol is the case in which a tailpipe-mass regulatory metric and a downwind sensor network disagree by construction.

doi: 10.1016/j.envres.2026.125881

Vocal fold fibroblasts 2026 Laryngoscope

Effects of Fine Particulate Matter ($PM_{2.5}$) in Cultured Human Vocal Fold Fibroblasts

METADATA ONLY. Metadata-only record: Wiley deposited title, authors and DOI on 7 Oct with no abstract. By title, an in-vitro exposure of human vocal fold fibroblasts to $PM_{2.5}$ - an unusual target tissue, and a plausible one given that the larynx sits in the deposition path of coarse and fine particles and that occupational voice disorders cluster in dusty trades. Relevance: low-moderate; on the retry list.

doi: 10.1002/lary.70976

Yadav et al. 2026 Environ Pollut

The dual role of atmospheric HULIS in inducing climate forcing and oxidative stress in the northwestern Himalayas

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 6 Oct with no abstract. By title, HULIS measured for both light absorption (brown-carbon forcing) and oxidative stress, in a high-altitude Himalayan setting. HULIS is the fraction that links the optical property a low-cost sensor responds to with the oxidative potential that toxicology implicates, so a single study reporting both is more useful than two reporting one each. Relevance: high; on the retry list.

doi: 10.1016/j.envpol.2026.129300

Zeng RON toxicity 2026 Sci Adv

Reduced organic nitrogen compounds dominate the toxicity of $PM_{2.5}$ organics following internal exposure

Nontargeted high-resolution MS of urban $PM_{2.5}$ organics is combined with protein-adductomics to define the fraction that actually reaches the circulation, then scored with multi-endpoint toxicity prediction. Reduced organic nitrogen (RON: amines, amides, N-heterocycles) carries 39% of the predicted multi-organ toxicity of the internally exposed fraction against 23% for the oxygenated organics the field usually targets, and binds human serum albumin strongly enough to explain systemic distribution. Limitation: the toxicity shares are model predictions on annotated features, not measured dose-response; nontargeted annotation confidence and ionisation bias toward basic nitrogen both inflate RON's apparent share. Relevance: high - the toxic fraction is invisible to optical mass sensing, an argument for adding composition proxies (e.g. N-containing BrC absorption) to networks.

doi: 10.1126/sciadv.aea8654

Occupational & indoor — 10 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (10 OF 10 RECORDS IN THIS SUBTOPIC)

Blackley et al. 2026 Ann Work Expo Health

Aerosol mitigation efficiency of high-volume suction systems during aerosol-generating dental procedures with patients

Five commercial high-volume suction systems - standard HVE, three intraoral systems and one modified HVE - measured during real crown preparations (n=26) and ultrasonic scaling (n=25), with direct-reading instruments near the dentist and in the hallway. Procedure-to-background concentration ratios were modelled with covariates (suction flow rate, procedure duration, tooth location, scaler type and quadrant); a system counted as effective only when the 95% lower confidence limit of the ratio was <=1, i.e. indistinguishable from background. Limitation: unblinded single-site evaluation with modest per-system n, patients differ across procedures, and the short-duration ratio design cannot speak to a full clinical day. Relevance: high - this is the design a sensing group should copy. The estimand is 'did the control return the air to background', tested against the instrument's own noise, rather than a percentage reduction quoted without a comparison.

doi: 10.1093/annweh/wxag081

Chen dormitory Stokes 2026 Build Environ

Size-resolved aerosol transport and exposure risk in university dormitories driven by Stokes number topology and source-flow coupling

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 9 Oct with no abstract. By title, size-resolved aerosol transport and exposure risk in dormitories organised around Stokes number and source-flow coupling - i.e. the regime where inertia, not the mean flow, decides where particles land, which is why a single room-average concentration misrepresents exposure for coarse particles. Relevance: moderate - the argument for size-resolved rather than $PM_{2.5}$-mass indoor nodes. On the abstract-retry list.

doi: 10.1016/j.buildenv.2026.115373

Kazakhstan PRISm 2026 Front Public Health

Occupational associations of preserved ratio impaired spirometry (PRISm) in a population-based study

2,275 adults from the general population with spirometry and a lifetime job-exposure matrix. PRISm prevalence was 12% (95% CI 10.7-13.3), higher in women (14%) than men (10%). Occupational airborne exposure including VGDF was not associated with PRISm, as ever-exposure or cumulative years; BMI was (adjusted OR 1.05 per unit). Limitation: a job-exposure matrix built for European job titles applied to Kazakh work histories misclassifies heavily and will null out a real association; cross-sectional design cannot separate PRISm from healthy-worker selection out of dusty jobs. Relevance: moderate - a clean negative in a heavily industrial country, and a direct argument for measured rather than matrix-assigned occupational exposure.

doi: 10.3389/fpubh.2026.1921126

Ketsakorn and Chaiyadej 2026 PLoS One

Probabilistic health risk assessment of respirable dust exposure among stone carvers in Thailand using average daily dose and Monte Carlo simulation

Personal sampling across mining, cutting, splitting, knocking and carving, with a 10,000-iteration Monte Carlo ILCR using US EPA exposure parameters. Mean respirable dust was below both the Thai OEL and the TLV everywhere, highest in carving at 0.051 mg/m3; ILCR stayed under 1e-6, highest in carving at 3.3e-7. Limitation: respirable dust mass is the wrong metric for a silica hazard - no crystalline silica fraction is reported, and an EPA inhalation-unit-risk framework applied to a mixed mineral dust produces a number with no established exposure-response behind it; small sample, one season. Relevance: low-moderate - the task-level contrast is exactly what a wearable logging monitor resolves and a single shift-average sample does not.

doi: 10.1371/journal.pone.0359706

Mahmoud filtration risk 2026 Build Environ

The link between aerosol dispersion, infection risk and air filtration

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 9 Oct with no abstract. By title, a link between indoor aerosol dispersion, filtration and infection risk - the Wells-Riley family of models, whose weak point is that a well-mixed room assumption and a measured clean-air delivery rate rarely both hold. Relevance: moderate - filtration intervention studies in this corpus (the 4 Oct AFRI-c HEPA null among them) keep failing on exposure verification, which is a monitoring problem before it is a filtration problem. On the abstract-retry list.

doi: 10.1016/j.buildenv.2026.115378

Neonatal air cleaners 2026 Build Environ

Portable air cleaners in neonatal facilities: Hybrid experimental-CFD assessment of different configurations

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 8 Oct with no abstract. By title, a combined experimental and CFD assessment of portable air cleaner configurations in neonatal facilities. Placement, not device rating, is what decides delivered clean-air rate in a real room, and the AFRI-c care-home null (4 October) is the clinical consequence of getting it wrong unmeasured. A neonatal unit is the setting where the dose argument is strongest. Relevance: high; on the retry list.

doi: 10.1016/j.buildenv.2026.115381

Othman et al. 2026 Indoor Air

Impact of Urban Land Use on Indoor Air Quality: Quantifying Ambient Particle Infiltration and Asthma Risk in Naturally Ventilated School Buildings

Ten primary schools stratified by land use (high-traffic, industrial, residential, mixed industrial/high-traffic, mixed industrial/port) with paired indoor-outdoor $PM_{2.5}$ and PM_1. Indoor maxima were at mixed industrial/port schools (24 h $PM_{2.5}$ 35.33 +/- 9.25, PM_1 23.79 +/- 5.87 $\mu g\,m^{-3}$); outdoor maxima at high-traffic schools (34.50 +/- 20.64 and 23.78 +/- 13.09); land-use class differences were significant ($p<0.05$) and the mean infiltration factor exceeded 0.6. Limitation: ten buildings across five classes is two per class, so the land-use contrast is confounded with building; Finf estimated from indoor-outdoor ratios without an independent air-change measurement absorbs any indoor source into the infiltration term. Relevance: high - Finf > 0.6 in naturally ventilated classrooms means outdoor network data is a usable proxy for children's exposure in this climate, which is the assumption most school studies make without testing.

doi: 10.1155/ina/5050009

Rees et al. 2026 NIHR Open Res

Lessons learned from using high efficiency particulate air (HEPA) filtration units in care homes: interviews with care home staff, residents and relatives from the AFRI-c trial

Reflexive thematic analysis of interviews with 25 staff, 20 residents and 12 relatives inside the AFRI-c pragmatic cluster RCT, whose main result was that HEPA units did not reduce respiratory infections. Units became normalised but were judged unsuitable for some residents; lockable settings and wall fixing were asked for; cost was seen as prohibitive sector-wide, raising an equity concern that filtration would reach only residents who can pay. Maintenance burden was minor. Limitation: qualitative, post hoc, and conducted among people who already knew the trial was running; no measurement of unit runtime, filter loading or achieved air-change rate - which is the missing variable that would explain a null. Relevance: high - a filtration null without exposure verification is the standing weakness of this trial literature, and a logging low-cost PM node in each room is the cheap fix.

doi: 10.3310/nihropenres.13993.1

Zhang WWTP bioaerosol 2026 Environ Res

Comprehensive Characterization of Bioaerosol Microbial Communities, ARG, and Occupational Exposure Risks in WWTP and Comprehensive Landfill

Paired sampling at a wastewater plant and an integrated solid-waste facility with 16S sequencing, HT-qPCR, QMRA and ML. Culturable bacteria were 3.4-fold higher at the landfill (1,726 vs 502 CFU/m3) and so was the fine biogenic particle share (27% vs 15%), but airborne ARG/MGE copies were an order of magnitude higher at the WWTP (5,462 vs 584 copies/m3), giving it the larger estimated worker inhalation dose. Limitation: two facilities, short campaigns, and culturable counts undercount viable-but-nonculturable cells; ARG copies per m3 do not establish viable resistant organisms. Relevance: low-moderate - optical sensors see biogenic particles as generic mass; the case for fluorescence-capable occupational monitors.

doi: 10.1016/j.envres.2026.125888

Zhao et al. 2026d Build Environ

Behavior-resolved bioaerosol exposure modeling reveals scale-dependent occupational risk in pediatric wards

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 5 Oct with no abstract. By title, a bioaerosol exposure model in paediatric wards resolved by occupant behaviour, reporting scale-dependent occupational risk. Behaviour-resolved models are the class of work that decides whether a fixed room monitor can stand in for a worker's exposure at all, and the paediatric ward is a setting where that substitution is routinely assumed. Relevance: potentially high; on the retry list.

doi: 10.1016/j.buildenv.2026.115364

Burden, policy & mitigation — 14 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (14 OF 14 RECORDS IN THIS SUBTOPIC)

Abdella Italy urban-rural 2026 Atmos Environ

Cleaning the air or shifting the burden? Characterizing urban-rural disparities in air pollution-attributable mortality across Italy

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 10 Oct with no abstract. By title, an attributable-mortality analysis asking whether Italian air-quality gains have been distributed or merely displaced from urban to rural populations. Rural exposure is systematically the worse-measured half of any national network, so a rural-urban disparity estimate inherits the monitoring asymmetry it is trying to describe - the 8 October US block-level record showed spatial aggregation attenuating exactly this kind of disparity. Relevance: moderate-high; on the abstract-retry list.

doi: 10.1016/j.atmosenv.2026.122420

BC outpatient risk 2026 Eco Environ Health

Black carbon as a major contributor to ambient $PM_{2.5}$-related cause-specific outpatient risks: implications for early warning and health benefits evaluation of air quality improvement

Over 212 million daily outpatient records, 2013-2017, with conditional logistic regression per city pooled by random effects and three complementary models testing constituent independence. Black carbon carried the strongest independent associations, and at lag0 rather than the lag0-1 typical of admissions: per IQR (1.8 $\mu g\,m^{-3}$) increases ranged from 1.33% (0.93-1.74) for chronic kidney disease to 3.89% (3.30-4.49) for COPD, attributable fractions 2.38% to 6.12%. Both $PM_{2.5}$- and BC-attributable burdens fell substantially after the 2013 clean-air programme. Limitation: constituents come from a modelled reanalysis and are collinear; outpatient visits are supply-sensitive, and the lag0 finding could reflect care-seeking on visibly bad days rather than a faster biological response. Relevance: high - BC is measurable by low-cost aethalometry, and a same-day constituent signal is the one an operational early-warning network could actually act on.

doi: 10.1016/j.eehl.2026.100283

China BC burden 2026 Nat Commun

The Hidden Health and Climate Burden of Black Carbon under China's Dual Carbon Strategy

A localized BC inventory and revised risk function put over 540,000 premature deaths attributable to black carbon in China in 2015 - more than twice previous estimates - while the CMIP6 inventory is found to overestimate BC radiative forcing by 45%. Projections suggest demographic ageing may offset the health gains from falling emissions. Limitation: a doubled burden rests on a 'revised health risk function' whose derivation is the entire result and cannot be checked from the abstract; BC-specific concentration-response functions are confounded with co-emitted species in every source study they come from, so attributing 540,000 deaths to BC rather than to combustion mixture is a strong claim. Relevance: high - BC is measurable by low-cost aethalometry, and the inventory-vs-measurement gap this paper quantifies is what a dense BC network would close.

doi: 10.1038/s41467-026-77632-8

Guo lung cancer burden 2026 Chin Med J Pulm Crit Care Med

Trends in lung cancer burden, risk factors, and disease projections in China and the United States, 1990-2023

GBD 2023 data with joinpoint regression, three-factor Das-Gupta decomposition and Bayesian age-period-cohort projection to 2050, comparing lung cancer burden and its attributable risks - smoking, air pollution, occupational exposure, metabolic and dietary factors - between China and the United States. Limitation: GBD attribution for air pollution is a modelled counterfactual built on exposure surfaces and borrowed concentration-response functions, so the pollution share is an assumption propagated forward, not an observation; the abstract reports no attributable fractions. Relevance: low-moderate - useful as the burden framing that the measurement side of this corpus feeds, but it adds no exposure information of its own.

doi: 10.1016/j.pccm.2026.08.006

Indo-Gangetic BC 2026 Atmos Pollut Res

Long-term decline in Black Carbon Aerosol and Associated Health Risk over north-western Indo-Gangetic Plain

METADATA ONLY. Metadata-only record: title, authors and DOI deposited 8 Oct with no abstract. By title, a long-term decline in black carbon over the north-western Indo-Gangetic Plain with an attached health-risk estimate. A declining BC trend in one of the most polluted regions on earth is a strong claim that depends entirely on aethalometer calibration and filter-loading correction being stable across the record - the commonest source of spurious long-term trends in BC. The same day carries two independent BC burden records (China, 540,000 deaths; the 274-city outpatient series on 7 October), so the trend direction matters. Relevance: high; on the retry list.

doi: 10.1016/j.apr.2026.103227

Izmir maritime 2026 Atmos Environ

Quantifying Maritime Contributions to Urban Air Quality in Izmir Bay Using Tier 3 Emission Inventory and AERMOD

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 8 Oct with no abstract. By title, a bottom-up Tier 3 (movement-based) ship emission inventory coupled to AERMOD for a port city. Port and shipping contributions are among the hardest urban sources to attribute from a land-based network, because the source is offshore, mobile and upwind only part of the time - the same problem the Geoje Island AMS record (6 October) attacked by measurement rather than by model. Relevance: moderate-high; on the retry list.

doi: 10.1016/j.atmosenv.2026.122410

Ma TRWP China 2026 Nat Commun

Spatiotemporal dynamics of tire wear particles reveal escalating emission inequalities across China's road network

A road-segment-resolved national inventory fuses harmonised field measurements, ML and traffic activity: China's roads emitted 425 kt of TRWP-derived $PM_{2.5}$ in 2023, ~80% from modelled vehicle-induced resuspension. TRWP's share of traffic $PM_{2.5}$ is projected to rise 34-fold from 2011 to 2050, reaching 67% of traffic $PM_{2.5}$ emissions, with exposure concentrated in megacity corridors; combined mitigation cuts emissions 41%. Limitation: resuspension dominates the total and is the least constrained term - it is modelled, not measured, and its emission factor varies by an order of magnitude with road silt loading and humidity. Relevance: high - non-exhaust PM is coarse-skewed and rubber/mineral-rich, exactly the aerosol on which Plantower-class sensors misreport mass, so a growing TRWP share silently changes network calibration.

doi: 10.1038/s41467-026-77475-3

Mukhamediyarov Kazakhstan code 2026 Atmosphere

Air Quality and Emission Trends Following the 2021 Environmental Code Reform in Kazakhstan Across Regions and Industrial Sectors

A before-and-after comparison of Kazakhstan's 2021 Environmental Code reform, which tightened standards and pushed Best Available Techniques. Reported stationary-source emissions fell roughly 5-6% between 2021 and 2024, but no statistically significant improvement in assessed ambient indicators appeared between 2021 and 2025, and episodic PM exceedances persisted in several cities. Limitation: no control region and no counterfactual, so the null is as much about the design as about the policy; reported emissions are self-declared inventory, and the ambient network is sparse enough that a 5% emission change would be undetectable even if real. Relevance: moderate - a clean example of why policy evaluation needs measurement density the regulatory network does not have; the 7 October congestion-pricing record is the version that had it.

doi: 10.3390/atmos17100993

NYC congestion pricing 2026 Environ Sci Technol Lett

Attributing $PM_{2.5}$ and NO_2 Changes One Year after New York City's Congestion Pricing Policy

The Central Business District Tolling Program began in January 2025; its first year is evaluated against 2022-2024 with difference-in-differences on NYCCAS real-time $PM_{2.5}$ and on satellite NO_2. Within the congestion relief zone, $PM_{2.5}$ fell by 1.44 $\mu g\,m^{-3}$ attributable to the policy ($p<0.001$), or 1.33 $\mu g\,m^{-3}$ allowing for spillover, while nearby non-zone sites showed a non-significant 0.50 $\mu g\,m^{-3}$ rise. Tropospheric NO_2 fell over 20% across metropolitan New York against a 2018-2024 baseline, but only 1.23% was attributable to the programme ($p=0.096$). Limitation: one treated zone and one post-period, so the parallel-trends assumption rests on three pre-years; the non-zone increase is the signal that displacement may be real but underpowered. Relevance: high - a 1.4 $\mu g\,m^{-3}$ step change across a few square kilometres is exactly the contrast a dense low-cost network resolves and a regulatory network, here a purpose-built survey, only barely does.

doi: 10.1021/acs.estlett.6c00610

Ranzi et al. 2026 Epidemiol Prev

Atlante Aria e Salute: methods for estimating the health impacts of air pollution and national application

Methodological description of the Air and Health Atlas, an interactive national platform. $PM_{2.5}$ and NO_2 are modelled at 1 km with random forests, converted to population-weighted exposure at municipality level against 2021 census blocks, and combined with provincial death-certificate rates through HRAPIE-2 concentration-response functions against the 5 $\mu g\,m^{-3}$ WHO counterfactual; 2019 is shown as an illustration, with pronounced north-south gradients. Limitation: population-weighted exposure at municipality scale cannot capture within-municipality contrast, the random-forest surface is validated against the same monitors that train it, and HRAPIE-2 functions are transferred from other populations. Relevance: high - this is the operational template a low-cost network would be asked to improve, and the 1 km population-weighted step is exactly where added density pays.

doi: 10.19191/ep26.4-5.s1.a1030.092

Rodriguez Bogota car-free 2026 Research Square (preprint)

Annual trends in ambient particulate matter during car free days in Bogota from 2016 to 2025

Nine annual Car-Free Day observations from the regulatory network, OLS and weighted regressions with 2,000-replicate bootstrap. PM_{10} fell significantly, slope -4.04 $\mu g\,m^{-3}$ per year (weighted bootstrap CI -6.49 to -0.81), median change -41.8%; $PM_{2.5}$ trend was -1.72 $\mu g\,m^{-3}$ per year (CI -3.17 to 0.40), not significant. Limitation: $n = 9$ single days with no control days, so the trend is the city's background trend plus that day's weather - it says nothing about the effect of the car-free intervention itself. Relevance: low-moderate - the right design (Car-Free Day minus matched control days, by station) is one that a dense sensor network could deliver at street scale.

doi: 10.21203/rs.3.rs-10864736/v1

US vehicle emission burden 2026 Environ Sci Technol

Spatially Refined Analysis of Emission Burdens and Equity from Light-, Medium-, and Heavy-Duty Vehicle Traffic in the United States

Block-level on-road emission intensities for the whole United States. Medium- and heavy-duty trucks are under 11% of vehicle kilometres but 46% of NOx and 50% of $PM_{2.5}$ emission burden; truck $PM_{2.5}$ is exhaust-dominated while tyre and brake wear are nearly half of light-duty $PM_{2.5}$. People of colour face burdens averaging 40% above the national mean (Asian populations over 60%), lower-income populations 20-25% higher, with significant disparities in 85-89% of counties, rural as well as urban. Critically, spatial aggregation attenuates the measured disparity. Limitation: this is an emission-intensity metric, not a concentration or an exposure - no dispersion, no background, no infiltration - so it cannot be read as an inhaled dose. Relevance: high - the aggregation result is the quantitative case for network density: resolution is not cosmetic, it determines whether a disparity is visible at all.

doi: 10.1021/acs.est.6c12227

Zhao et al. 2026b Atmosphere

Pollution Characteristics, Influencing Factors and Health Risk Assessments of Mercury in Atmospheric $PM_{2.5}$ from Xinxiang, China

Year-long filter sampling gives an annual mean $PM_{2.5}$-bound Hg of 0.399 ng/m3, significantly higher in winter ($p<0.001$) and in the heating than the non-heating period, positively correlated with SO_2, CO, $PM_{2.5}$, OC, EC, Cl-, Zn and Ca and negatively with temperature, O_3 and V - a coal-combustion signature. Hazard index 2.06e-3 (children) and 8.46e-4 (adults), both far below 1, with ingestion the dominant modelled pathway. Limitation: the samples are from 2015 and describe an emissions regime two major control phases out of date; an ingestion-dominant pathway for an airborne metal is an artefact of the EPA soil-oriented default parameters. Relevance: low - logged as a composition time point, not as a risk estimate.

doi: 10.3390/atmos17100974

Zhao et al. 2026e Research Square (preprint)

Thermal-oxygen evolution and atmospheric pollutant emission characteristics during the vertical propagation of underground coal fires

A scaled physical-similarity rig run from ignition to burnout with continuous monitoring at six depths. Thermal-oxygen evolution split into incubation, rapid heating, intensive reaction, rapid cooling and burnout; the upper layers burned out while the middle sustained reaction and the deepest was oxygen-limited; surface response lagged the seam. Particulate matter and light hydrocarbons were released early (CH4 and C2H6 concentrated in the first 40-60 h, at background by ~80 h) while CO_2 and CO dominated later. Limitation: a similarity-scaled rig reproduces sequence, not magnitude - emission factors from it cannot be transferred to a real coal fire, and no scaling validation is offered; preprint. Relevance: moderate - the early-PM, late-CO sequence is a usable detection signature for surface sensor arrays over suspected underground fires.

doi: 10.21203/rs.3.rs-10517753/v1

Other clinical endpoints — 4 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (4 OF 4 RECORDS IN THIS SUBTOPIC)

Beijing CKD cohort 2026 Environ Pollut

Association between Long-term Exposure to Air Pollution and Incident Chronic Kidney Disease among Adults in Beijing, China

15,580 adults in the Beijing Air Pollution and Population Health Cohort, 664 incident CKD cases 2019-2025, time-varying Cox models. Per 10 $\mu g\,m^{-3}$: PM_{10} HR 1.18 (1.06-1.31) and CO 1.02 (1.01-1.03) were significant; $PM_{2.5}$ 1.22 (0.99-1.50) and NO_2 1.17 (0.96-1.43) were positive but crossed unity; O_3 was inverse, 0.77 (0.68-0.87). The composite score gave 1.51 (1.07-2.12) per IQR, with stronger associations in women, older adults and urban residents. Limitation: six years of follow-up in one city where all pollutants fell together makes single-pollutant HRs nearly uninterpretable - the inverse ozone estimate is the tell, since ozone rose as PM fell; eGFR-defined incident CKD is sensitive to a single creatinine measurement. Relevance: moderate - kidney endpoints are under-studied, but this design cannot separate the pollutants.

doi: 10.1016/j.envpol.2026.129302

CLHLS indoor constituents 2026 BMC Public Health

Association between exposure to indoor $PM_{2.5}$ constituents and mortality risk among Chinese older adults: a nationwide prospective cohort study

12,607 participants (mean age 86.8) followed 2008-2018, 8,643 deaths over a median 4.14 years, time-varying Cox with restricted cubic splines. Per 1 $\mu g\,m^{-3}$: $PM_{2.5}$ HR 1.003 (1.001-1.005), black carbon 1.072 (1.035-1.111), organic matter 1.009 (1.002-1.016), sulfate 1.010 (1.001-1.019); associations were stronger in rural residents, non-linear, robust to excluding hypertensive and diabetic participants, and $PM_{2.5}$ tracked the neutrophil-to-lymphocyte ratio as a candidate inflammatory mediator. Limitation: 'indoor' here is an ambient field multiplied by a modelled infiltration factor, not a measurement - and the infiltration factor is itself a function of building and season, so constituent contrasts partly encode housing type; black carbon's HR is an order of magnitude above $PM_{2.5}$'s per unit mass, which is biologically plausible and also what collinearity with a small-magnitude constituent produces. Relevance: high - this is the exact quantity a cheap indoor node measures directly, and the study had to model it.

doi: 10.1186/s12889-026-29130-1

Hefei conjunctivitis 2026 Front Public Health

Ambient air pollutants and outpatient visits for conjunctivitis in Hefei, China: a time-series study of hospital-based data during 2014-2023

73,043 conjunctivitis outpatient records, 2014-2023, in a time-series design with stratification by sex, age, season and pandemic period. All five pollutants were positively associated with daily visits, strongest generally at lag0; SO_2 carried the largest lag-0 estimate, then NO_2, with pollutant-specific cumulative lag patterns. Limitation: no effect sizes, units or intervals appear in the abstract, so nothing can be entered; SO_2 ranking first in a city where SO_2 has fallen to a few $\mu g\,m^{-3}$ suggests a marker for combustion-season conditions rather than a conjunctival effect, and single-city time series cannot separate co-varying pollutants. Relevance: low-moderate - ocular surface endpoints are plausible for coarse and irritant exposure, and a record of the claim is worth keeping even though this analysis does not establish it.

doi: 10.3389/fpubh.2026.1926405

Tian et al. 2026 Hum Genet

Annual average exposure to ambient air pollution, polygenic risk, and incidence of major chronic diseases: a large-scale prospective cohort study

313,534 UK Biobank participants with LUR annual pollutant estimates and weighted polygenic risk scores for each of five disease groups; Cox models with RERI and AP for additive interaction. $PM_{2.5}$ carried the strongest association, HR 1.22 per 5 $\mu g\,m^{-3}$ (95% CI 1.19-1.26); high genetic risk acted independently (diabetes HR 1.77 per SD of PRS), with significant additive interaction between high PRS and high $PM_{2.5}$ for neurodegenerative disease (RERI 0.14) and diabetes (RERI 0.21). Limitation: UK Biobank is a healthy-volunteer cohort with a narrow $PM_{2.5}$ range, the LUR surface is a single baseline-era estimate carried forward, and five outcomes with four pollutants invites selective emphasis - the interaction terms are modest and not corrected for multiplicity. Relevance: moderate - gene-environment interaction raises the value of resolving exposure well at the individual level, which is the case for personal rather than ambient monitoring.

doi: 10.1007/s00439-026-02875-w

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
ABCD air pollution	The Relationship between Air Pollution Exposure, Genetic Liability for Schizophrenia, and Brain Outcomes in the ABCD Study	*Schizophr Bull*	Prospective cohort	Modelled annual $PM_{2.5}$, NO2 and O3 at age 9-10	USA (ABCD Study)
Batisse et al.	Long-term exposure to ultrafine particles and mortality from neurodegenerative diseases: a population-based cohort study	*Epidemiology*	Retrospective cohort	High-resolution UFP number concentration and UFP size at residential postal code, 3-year moving average	Canada (Montreal and Toronto)
Byun ADRD readmission	$PM_{2.5}$ and 30-Day Readmission in Medicare Beneficiaries with Alzheimer's Disease and Related Dementias: Effect Modification by Neighborhood Green and Blue Space	*Environ Health Perspect*	Time-stratified case-crossover	Daily modelled $PM_{2.5}$ at residential ZIP Code, lag 0-2	United States (contiguous, Medicare 2000-2016)
$PM_{2.5}$ and Mn biomarkers	Differential associations of $PM_{2.5}$ and particulate manganese with blood-based amyloid and tau biomarkers	*Environ Pollut*	Metadata only (no abstract)	Metadata only - $PM_{2.5}$ and particulate manganese against blood amyloid and tau biomarkers (by title)	Not recoverable from metadata
Tang source PM dementia	Associations of long-term exposure to source-specific $PM_{2.5}$, lifestyle, and genetic susceptibility with dementia risk: a prospective cohort study	*Eur Arch Psychiatry Clin Neurosci*	Prospective cohort	Source-specific $PM_{2.5}$ (domestic, road transport, other transport, industry), time-varying	Not stated (large prospective cohort)
Wang et al.	Long-Term Air Pollution Exposure and Brain Structural Decline: A Longitudinal Analysis	*Environ Pollut*	Prospective cohort	Residential $PM_{2.5}$ and PM_{10} from high-resolution spatiotemporal models	Not stated in abstract (middle-aged and older adult imaging cohort)
Cardiovascular & metabolic					
Children's AIRE	Ambient particulate matter exposure and children's blood pressure in rural California, 2019-2022	*Int J Hyg Environ Health*	Prospective cohort	12-month and 7-day $PM_{2.5}$ and PM_{10} from regulatory monitors linked to residence	USA (rural California)
Reproductive & developmental					
Qiang maternal PM macrosomia	Interaction of maternal $PM_{2.5}$ exposure and psychosocial state on newborn macrosomia: A nationwide population-based study	*Chin Med J (Engl)*	Metadata only (no abstract)	Metadata only - maternal $PM_{2.5}$ exposure with psychosocial state, nationwide (by title)	China (nationwide)
Respiratory & allergic					
Howlett-Downing and Wichmann	Triangulating health risks of $PM_{2.5}$ and trace elements in Pretoria: evidence from health risk assessment, case-crossover, and concentration-response function analyses	*Int J Environ Health Res*	Time-stratified case-crossover	Sampled $PM_{2.5}$ mass and $PM_{2.5}$-bound trace elements, Apr 2017 - Feb 2020	South Africa (Pretoria)
Ko-CHENS 36 months	Exposure to Environmental Pollutants and the Risk of Respiratory Illnesses in 36-Month-Old Children: The Korean Children's Environmental Health Study	*Allergy Asthma Immunol Res*	Prospective birth cohort (mixtures)	Home-visit indoor PM_{10} and $PM_{2.5}$ with VOCs and dust-mite allergen; blood metals and urinary metabolites	South Korea (Ko-CHENS)
Sao Paulo paediatric	Climate and air pollution are associated with pediatric respiratory hospitalizations in Sao Paulo, Brazil	*Front Pediatr*	Observational - cross-sectional	IDW-interpolated pollutant and meteorological fields at geocoded residence, 2022	Brazil (Metropolitan Sao Paulo)

First author	Focus	Journal	Design	Metric	Region
Mechanistic toxicology					
Aged traffic particles	Aging of Urban Traffic Fine Particles Modulates Toxicity Mechanisms and Oxidative Stress Responses	*Environ Sci Technol*	Chamber / laboratory	Parking-garage aerosol (cold-start plus urban), fresh and OFR-aged; oxidative potential, SOA mass, O:C	Not stated in abstract (parking-garage sampling)
Env epigenetics review	Environmental epigenetic modifiers in neurodegenerative diseases: a systematic review of epigenomic mechanisms and therapeutic targets	*Front Epigenet Epigenom*	Systematic review (PRISMA)	Particulate matter among heavy metals, gaseous pollutants and synthetic neurotoxicants (review)	Global
Gao et al.	Air pollution and early-onset lung cancer: from environmental carcinogenesis to early intervention strategies	*Carcinogenesis*	Narrative review	Ambient and household $PM_{2.5}$/PM_{10} with PAH, metal and VOC components (review)	Global
Melatonin $DPM_{2.5}$	Melatonin protects keratinocytes from diesel-derived $PM_{2.5}$-induced damage by suppressing JNK/p38 MAPK signaling and caspase activation	*Ecotoxicol Environ Saf*	Animal + in vitro	Diesel-derived $PM_{2.5}$ applied to HaCaT keratinocytes and to HR-1 hairless mice	South Korea (laboratory)
Mountain West wood smoke	Household wood smoke exposure, lung black carbon deposition, and genomic instability in U.S. Mountain West residents	*Part Fibre Toxicol*	Metadata only (no abstract)	Metadata only - household wood smoke, lung black carbon deposition and genomic instability (by title)	USA (Mountain West)
Sharma et al.	Characterization of carbonaceous particles generated from three flame modes using a single burner to enable properties-based exposure studies	*Aerosol Sci Technol*	Metadata only (no abstract)	Metadata only - carbonaceous particles from three flame modes on one burner (by title)	Not recoverable from metadata
Tampere SOA toxicity	From primary to secondary aerosol: chemical transformation and toxicity of gasoline vehicle exhaust under contrasting driving conditions	*Environ Res*	Chamber / laboratory	Primary and photochemically aged gasoline exhaust; PAHs, nitro- and oxy-PAHs, water-soluble elements, SP-AMS	Finland (Tampere, chassis dynamometer)
Vocal fold fibroblasts	Effects of Fine Particulate Matter ($PM_{2.5}$) in Cultured Human Vocal Fold Fibroblasts	*Laryngoscope*	Metadata only (no abstract)	Metadata only - $PM_{2.5}$ applied to cultured human vocal fold fibroblasts (by title)	Not recoverable from metadata
Yadav et al.	The dual role of atmospheric HULIS in inducing climate forcing and oxidative stress in the northwestern Himalayas	*Environ Pollut*	Metadata only (no abstract)	Metadata only - humic-like substances, light absorption and oxidative potential (by title)	India (northwestern Himalayas)
Zeng RON toxicity	Reduced organic nitrogen compounds dominate the toxicity of $PM_{2.5}$ organics following internal exposure	*Sci Adv*	Chamber / laboratory	Urban $PM_{2.5}$ organic fraction: nontargeted HRMS, serum-albumin adductomics, multi-endpoint toxicity prediction	China (urban)
Occupational & indoor					
Blackley et al.	Aerosol mitigation efficiency of high-volume suction systems during aerosol-generating dental procedures with patients	*Ann Work Expo Health*	Field measurement	Direct-reading number and mass concentrations, PM4 (respirable) and PM_{10} (thoracic), with size distribution	USA (NIOSH, clinical dental operatories)

First author	Focus	Journal	Design	Metric	Region
Chen dormitory Stokes	Size-resolved aerosol transport and exposure risk in university dormitories driven by Stokes number topology and source-flow coupling	*Build Environ*	Metadata only (no abstract)	Metadata only - size-resolved indoor aerosol transport and exposure, Stokes-number framing (by title)	Not stated (university dormitories)
Kazakhstan PRISm	Occupational associations of preserved ratio impaired spirometry (PRISm) in a population-based study	*Front Public Health*	Observational - cross-sectional	Lifetime occupational vapour, gas, dust and fume exposure by job-exposure matrix; no PM measured	Kazakhstan (Aktobe, Karaganda, Kostanay)
Ketsakorn and Chaiyadej	Probabilistic health risk assessment of respirable dust exposure among stone carvers in Thailand using average daily dose and Monte Carlo simulation	*PLoS One*	Observational - cross-sectional	Personal respirable dust, NIOSH 7601, 8-h TWA across five task groups	Thailand
Mahmoud filtration risk	The link between aerosol dispersion, infection risk and air filtration	*Build Environ*	Metadata only (no abstract)	Metadata only - indoor aerosol dispersion against filtration and modelled infection risk (by title)	Not stated (indoor)
Neonatal air cleaners	Portable air cleaners in neonatal facilities: Hybrid experimental-CFD assessment of different configurations	*Build Environ*	Metadata only (no abstract)	Metadata only - portable air cleaner placement assessed by experiment plus CFD (by title)	Not recoverable from metadata
Othman et al.	Impact of Urban Land Use on Indoor Air Quality: Quantifying Ambient Particle Infiltration and Asthma Risk in Naturally Ventilated School Buildings	*Indoor Air*	Field measurement	Paired indoor-outdoor 24 h $PM_{2.5}$ and PM1 with infiltration factor, 10 schools across five land-use classes	Malaysia (Klang Valley)
Rees et al.	Lessons learned from using high efficiency particulate air (HEPA) filtration units in care homes: interviews with care home staff, residents and relatives from the AFRI-c trial	*NIHR Open Res*	Qualitative study (within RCT)	HEPA filtration units as the intervention; no PM measured in this component	United Kingdom (care homes, AFRI-c cluster RCT)
Zhang WWTP bioaerosol	Comprehensive Characterization of Bioaerosol Microbial Communities, ARG, and Occupational Exposure Risks in WWTP and Comprehensive Landfill	*Environ Res*	Field measurement	Culturable bioaerosol (CFU/m3), fine biogenic particle fraction, airborne ARG/MGE copies	China (central)
Zhao et al.	Behavior-resolved bioaerosol exposure modeling reveals scale-dependent occupational risk in pediatric wards	*Build Environ*	Metadata only (no abstract)	Metadata only - behaviour-resolved bioaerosol exposure model in hospital wards (by title)	China (by journal and author affiliation; not asserted)
Sensing, forecasting & instrumentation					
Calabar ZigBee WSN	Design and Performance Evaluation of a ZigBee-Based Wireless Sensor Network for CO and $PM_{2.5}$ Air Quality Monitoring in Calabar Metropolis	*Research Square (preprint)*	Instrument development	Low-cost CO and $PM_{2.5}$ nodes on an IEEE 802.15.4 ZigBee mesh, with MATLAB models of sensing, link, energy and dispersion	Nigeria (Calabar)
Chang et al.	Wintertime vertical stratification of air pollutants and meteorological associations within an urban forest canopy in Shenyang, China	*Environ Pollut*	Long-term field measurement campaign	$PM_{2.5}$, PM_{10} with O3, NO2, SO2, CO and meteorology at 1.5, 10 and 20 m	China (Shenyang)
Chiappini WRF-Chem nudging	Regime-Dependent Effectiveness of Spectral Nudging in WRF-Chem: Meteorological and Saharan Dust Skill over the Central Mediterranean	*Atmosphere*	Chemical transport model	Modelled Saharan dust AOD550 against AERONET; meteorological RMSE from nudged vs free runs	Europe (Central Mediterranean)

First author	Focus	Journal	Design	Metric	Region
Dai et al.	Cross-phase partitioning of sulfur-nitrogen ratios and aerosol mixing-state evolution based on single-particle observations	*Atmos Chem Phys*	Long-term field measurement campaign	Single-particle aerosol mass spectrometry with gas-, particle- and number-based S/N ratios	China (Yangzhou, eastern China)
Dayare Navi Mumbai	Weather, Co-Pollutants and Daily $PM_{2.5}$ in a Coastal Indian City: Multi-Station Evidence for Meteorology-Triggered Air-Quality Warnings in Navi Mumbai	*Preprints.org (preprint)*	Monitoring-record analysis	Daily $PM_{2.5}$ at five CAAQMS stations (2021-2025), hourly at two; ERA5 checked against radiosonde	India (Navi Mumbai)
Fang LCS calibration review	Calibration of Low-Cost Particulate Matter Sensors for Exposure Assessment: A Critical Review	*Curr Environ Health Rep*	Systematic review (PRISMA)	Calibrated R2 and RMSE of low-cost PM sensors against reference instruments, laboratory vs field	Global (66 studies, 2015-2025)
Ghajari fuzzy DL Tehran	A Novel Fuzzy Hybrid Deep Learning Architecture for Spatiotemporal $PM_{2.5}$ Prediction	*Research Square (preprint)*	Machine-learning model	Station $PM_{2.5}$ with meteorological, satellite, topographic, population and road inputs, 2014-2016	Iran (Tehran)
Hazelwood et al.	Size-Resolved Particle Dry Deposition Velocities in a Controlled Spherical Chamber: Effects of Turbulence, Surface Complexity, and Particle Physicochemical Properties	*J Aerosol Sci*	Metadata only (no abstract)	Metadata only - size-resolved dry deposition velocity in a controlled chamber (by title)	Not recoverable from metadata
Loh et al.	Seasonal characteristics of submicron aerosols in the coastal atmosphere of southeastern Korea: Influence of urban-port outflow	*Mar Pollut Bull*	Long-term field measurement campaign	PM1 mass, composition and size distribution by HR-ToF-AMS, four seasons, with PMF	South Korea (Geoje Island, downwind of Busan)
Malik MPSS calibration	Traceable calibration of MPSS in fast-scan mode: Recommendations for PNSD measurement at urban traffic hot spots	*Aerosol Sci Technol*	Metadata only (no abstract)	Metadata only - traceable calibration of mobility particle size spectrometers in fast-scan mode (by title)	Not stated (calibration method)
Mei et al.	Synthesis of the tethered balloon system and other TRACER campaign measurements elucidates aerosol property profiles	*Atmos Chem Phys*	Long-term field measurement campaign	Vertical profiles of aerosol number concentration and size distribution; CCN inferred by kappa-Koehler	USA (Houston, Texas)
Metris and Metris	Fungus above us: exploring the atmospheric mycobiome by aircraft	*PeerJ*	Field measurement	Airborne fungal bioaerosol, aircraft sampling 2022-2023 with remote-sensing and meteorological covariates	Europe (aircraft surveys; United Kingdom-based)
Pavlidis cold start	Measurements of Real-World Cold Start Emissions in a Parking Garage	*Environ Sci: Atmos*	Field measurement	Real-world cold-start particle and gas emissions measured in a parking garage	Not stated (urban parking garage)
Pires quantum NN	An Annealing-based Quantum Neural Network with Discrete Wavelet Transforms for Enhanced Air Pollution Nowcasting	*arXiv (preprint)*	Machine-learning model	Urban $PM_{2.5}$ time series (UK, high-density data)	United Kingdom
Zhang et al.	Full field of view temporal calibration for the directional polarimetric camera onboard the GaoFen-5 (02) satellite based on a data reconstruction method	*Appl Opt*	Metadata only (no abstract)	Metadata only - on-orbit temporal calibration of a directional polarimetric camera (by title)	China (GaoFen-5 (02) mission)
Exposure assessment & modelling					
Aerosol Fe isotopes	Isotopic composition of aerosol iron from anthropogenic sources: implications for source apportionment of aerosol iron	*Atmos Chem Phys*	Source apportionment	delta-56Fe end-members for desert dust and anthropogenic aerosol iron sources	Global (source characterisation)

First author	Focus	Journal	Design	Metric	Region
Anwar Pretoria FTIR	$PM_{2.5}$ Composition and Organic Matter Sources in Pretoria, South Africa Employing Fourier Transform-Infrared Spectroscopy	*Aerosol Air Qual Res*	Metadata only (no abstract)	Metadata only - FT-IR functional-group speciation of $PM_{2.5}$ organic matter (by title)	South Africa (Pretoria)
Arora rural PAH	Seasonal PAH Exposure and Health Risks at Central European Rural Background Sites Driven by Combustion Sources and Transport Pathways	*Atmos Environ*	Metadata only (no abstract)	Metadata only - particle-bound PAH at Central European rural background sites, with risk estimates (by title)	Europe (Central, rural background)
Balabat et al.	Urban Road Dust as a Microbial Hotspot: Spatial Variation of Bacterial Communities and Metal Associations in the Philippines	*World J Microbiol Biotechnol*	Field measurement	Road dust (resuspendable coarse reservoir): 16S V3-V4 metabarcoding, trace metals, pollution indices	Philippines (Metro Manila, Cebu, Cagayan de Oro)
Budapest deposition	Aerosol Deposition in the Human Respiratory System from Distinct Urban Sources Using Multiple Exposure Metrics	*Environ Sci Technol*	Exposure model (secondary data)	Source-apportioned particle number size distributions, 6-1000 nm, with lung deposition modelling	Hungary (Budapest urban background)
Carnec adjoint nodal IAQ	Adjoint-based corrective approach for nodal indoor air quality predictions	*Indoor Environments*	Metadata only (no abstract)	Metadata only - adjoint correction of nodal (multi-zone) indoor air quality models (by title)	Not stated (building model)
Chaudhary India meteorology	Meteorological modulation of ground-level air quality in India based on in situ measurements	*Environ Monit Assess*	Monitoring-record analysis	PM_{10}, $PM_{2.5}$, O3, NO2, SO2 with PBL height, RH, temperature, precipitation, wind, surface pressure (2017-2024)	India (national, Indo-Gangetic Plain focus)
Chavan and Lataye	Application of Openair package in R programming and PCA: unveiling air quality dynamics and meteorological influences	*Environ Sci Pollut Res Int*	Monitoring-record analysis	2023 CAAQMS records: PM_{10}, $PM_{2.5}$ with six co-pollutants and four meteorological variables	India (Nagpur)
Conformal $PM_{2.5}$ mapping	Preregistered External Validation of Spatially Valid Conformal Prediction for Air-Pollution Exposure Mapping in China	*Research Square* (*preprint*)	Statistical model	Province-level population-weighted $PM_{2.5}$, 1998-2024 (n=837), with distribution-free prediction intervals	China (provincial)
Fazelvalipour LA microplastics	Size-Resolved Characteristics and Source Influences of Airborne Microplastics in Central Los Angeles	*Preprints.org* (*preprint*)	Field measurement	PM_{10}-associated airborne microplastics (counts/m3, morphology, size) with PM_{10} composition	United States (central Los Angeles)
GOES-R $PM_{2.5}$ CONUS	Machine Learning Predictions of $PM_{2.5}$ and Their Applications to Exposure and Health Assessment in the CONUS	*GeoHealth*	Machine-learning model	Hourly $PM_{2.5}$ at 0.01 degrees from gap-filled GOES-R geostationary AOD plus random forest	USA (contiguous)
Hayli Gubbi eruption	Short-term Atmospheric Impacts of the Hayli Gubbi Volcanic Eruption 2025	*Atmos Environ*	Metadata only (no abstract)	Metadata only - short-term atmospheric impacts of a 2025 eruption (by title)	Ethiopia (Hayli Gubbi, Afar)
Hopke et al.	Characterization of PM Pollution Episodes in Kuwait	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - PM episode characterisation (by title)	Kuwait

First author	Focus	Journal	Design	Metric	Region
Mukhamediyarov et al.	Elemental Composition, Sources, and Health Risks of Ambient $PM_{2.5}$ in Ust-Kamenogorsk (Oskemen) and Pavlodar, Kazakhstan	*Atmosphere*	Source apportionment	$PM_{2.5}$ elemental composition by ICP-OES and ICP-MS with PCA source resolution	Kazakhstan (Ust-Kamenogorsk, Pavlodar)
N-containing OA	Unveiling the Precursor-Specific Aqueous Formation Mechanisms of Nitrogen-Containing Organic Aerosols	*Atmos Environ*	Metadata only (no abstract)	Metadata only - precursor-specific aqueous formation of nitrogen-containing organic aerosol (by title)	Not recoverable from metadata
$PM_{2.5}$ and GPP	Climate-dependent nonlinear relationships between $PM_{2.5}$ and ecosystem productivity across China's urban agglomerations	*Environ Pollut*	Machine-learning model	Satellite $PM_{2.5}$ against gross primary productivity across five urban agglomerations	China (five urban agglomerations)
Petrovic Belgrade PMF	Chemical characterization and source apportionment of PM_{10} in Belgrade, Serbia: influence of local and regional anthropogenic and natural sources	*Atmos Chem Phys*	Source apportionment	PM_{10} with 34 species (OC/EC, elements, ions, organic tracers), PMF 5.0	Serbia (Belgrade urban background)
Rocha Volta Redonda	Long-term trends and seasonal variability of particulate matter (PM_{10} and $PM_{2.5}$) in Volta Redonda (RJ), Brazil: a 15-year temporal analysis	*Environ Monit Assess*	Metadata only (no abstract)	Metadata only - 15-year PM_{10} and $PM_{2.5}$ record at a Brazilian steel-industry city (by title)	Brazil (Volta Redonda, RJ)
Shi et al.	Compositional transition and coupled formation pathways of $PM_{2.5}$ pollution in a typical industrial city: An empirical study based on Handan, China	*Environ Pollut*	Metadata only (no abstract)	Metadata only - $PM_{2.5}$ compositional transition and coupled formation pathways (by title)	China (Handan, Hebei)
Wang et al.	Physics-inspired inductive spatiotemporal kriging for $PM_{2.5}$ with satellite gradient constraints	*Atmos Environ*	Metadata only (no abstract)	Metadata only - inductive spatiotemporal kriging for $PM_{2.5}$ under satellite gradient constraints (by title)	Not recoverable from metadata
Wu NCP brown carbon	Light absorption properties and composition of water-soluble brown carbon in North China Plain: implication for an enhancing role of nitrogenous organic compounds	*Atmos Chem Phys*	Field measurement	Water-soluble BrC absorption (Abs365, MAE) and composition at five sites, winter 2023	China (North China Plain, five sites)
Yadav ENSO India	Air Quality in Flux: The Baffling ENSO Transition in India	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - ENSO-phase modulation of Indian air quality (by title)	India (national)
Zhao et al.	Spatial Heterogeneity and Multi-scale Statistical Associations of Urban Air Pollution Revealed by Interpretable Machine Learning in Shenzhen, China	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - multi-scale spatial heterogeneity via interpretable ML (by title)	China (Shenzhen)
Burden, policy & mitigation					
Abdella Italy urban-rural	Cleaning the air or shifting the burden? Characterizing urban-rural disparities in air pollution-attributable mortality across Italy	*Atmos Environ*	Metadata only (no abstract)	Metadata only - urban-rural disparity in PM-attributable mortality across Italy (by title)	Italy (national)
BC outpatient risk	Black carbon as a major contributor to ambient $PM_{2.5}$-related cause-specific outpatient risks: implications for early warning and health benefits evaluation of air quality improvement	*Eco Environ Health*	Time-stratified case-crossover	Daily $PM_{2.5}$ and constituents (black carbon, organic matter, sulfate, nitrate, ammonium), 274 cities	China (274 cities)
China BC burden	The Hidden Health and Climate Burden of Black Carbon under China's Dual Carbon Strategy	*Nat Commun*	Health impact model	Localized black carbon emission inventory with revised health risk function; CMIP6 comparison	China (national)

First author	Focus	Journal	Design	Metric	Region
Guo lung cancer burden	Trends in lung cancer burden, risk factors, and disease projections in China and the United States, 1990-2023	*Chin Med J Pulm Crit Care Med*	Health impact model	GBD 2023 risk-factor attribution including ambient and household air pollution	China and United States
Indo-Gangetic BC	Long-term decline in Black Carbon Aerosol and Associated Health Risk over north-western Indo-Gangetic Plain	*Atmos Pollut Res*	Metadata only (no abstract)	Metadata only - long-term black carbon trend and associated health risk (by title)	India (north-western Indo-Gangetic Plain)
Izmir maritime	Quantifying Maritime Contributions to Urban Air Quality in Izmir Bay Using Tier 3 Emission Inventory and AERMOD	*Atmos Environ*	Metadata only (no abstract)	Metadata only - Tier 3 ship emission inventory with AERMOD dispersion (by title)	Turkiye (Izmir Bay)
Ma TRWP China	Spatiotemporal dynamics of tire wear particles reveal escalating emission inequalities across China's road network	*Nat Commun*	Modelling / inventory	Tire and road wear particle (TRWP) $PM_{2.5}$ emissions, road-network resolved, ML-harmonised field data	China (national road network)
Mukhamediyarov Kazakhstan code	Air Quality and Emission Trends Following the 2021 Environmental Code Reform in Kazakhstan Across Regions and Industrial Sectors	*Atmosphere*	Monitoring-record analysis	Stationary-source emissions ($PM_{2.5}$, SO2, CO, NO2) and ambient indicators by region and sector	Kazakhstan (national)
NYC congestion pricing	Attributing $PM_{2.5}$ and NO2 Changes One Year after New York City's Congestion Pricing Policy	*Environ Sci Technol Lett*	Difference-in-differences evaluation	Real-time $PM_{2.5}$ from the NYC Community Air Survey plus TROPOMI tropospheric NO2 columns	USA (New York City)
Ranzi et al.	Atlante Aria e Salute: methods for estimating the health impacts of air pollution and national application	*Epidemiol Prev*	Health impact model	Modelled annual $PM_{2.5}$ and NO2 at 1x1 km, random forest over monitors, satellite and covariates, 2016-2024	Italy (national, municipality level)
Rodriguez Bogota car-free	Annual trends in ambient particulate matter during car free days in Bogota from 2016 to 2025	*Research Square (preprint)*	Monitoring-record analysis	$PM_{2.5}$ and PM_{10} on Car-Free Day from the city monitoring network, 2016-2025 (2021 excluded)	Colombia (Bogota)
US vehicle emission burden	Spatially Refined Analysis of Emission Burdens and Equity from Light-, Medium-, and Heavy-Duty Vehicle Traffic in the United States	*Environ Sci Technol*	Modelling / inventory	Census-block NOx and $PM_{2.5}$ emission intensity from MOVES factors and street-level traffic, exhaust vs tyre/brake wear	USA (national, every census block)
Zhao et al.	Pollution Characteristics, Influencing Factors and Health Risk Assessments of Mercury in Atmospheric $PM_{2.5}$ from Xinxiang, China	*Atmosphere*	Source apportionment	$PM_{2.5}$-bound mercury with OC/EC, water-soluble ions and trace elements, full-year 2015 sampling	China (Xinxiang, Henan)
Zhao et al.	Thermal-oxygen evolution and atmospheric pollutant emission characteristics during the vertical propagation of underground coal fires	*Research Square (preprint)*	Chamber / laboratory	$PM_{2.5}$, PM_{10} and TSP with CO2, CO, CH4 and C2H6 through a simulated burn cycle	China (laboratory, physical similarity simulation)
Other clinical endpoints					
Beijing CKD cohort	Association between Long-term Exposure to Air Pollution and Incident Chronic Kidney Disease among Adults in Beijing, China	*Environ Pollut*	Prospective cohort	Annual PM_{10}, $PM_{2.5}$, NO2, CO and O3, with a composite pollution score	China (Beijing)

First author	Focus	Journal	Design	Metric	Region
CLHLS indoor constituents	Association between exposure to indoor $PM_{2.5}$ constituents and mortality risk among Chinese older adults: a nationwide prospective cohort study	*BMC Public Health*	Prospective cohort	Indoor $PM_{2.5}$ and constituents (BC, OM, sulfate, ammonium, nitrate) from ambient fields x infiltration factors	China (nationwide, CLHLS-type cohort)
Hefei conjunctivitis	Ambient air pollutants and outpatient visits for conjunctivitis in Hefei, China: a time-series study of hospital-based data during 2014-2023	*Front Public Health*	Monitoring-record analysis	Daily $PM_{2.5}$ and PM_{10} with O3, NO2 and SO2 from regulatory monitors	China (Hefei)
Tian et al.	Annual average exposure to ambient air pollution, polygenic risk, and incidence of major chronic diseases: a large-scale prospective cohort study	*Hum Genet*	Prospective cohort	Annual $PM_{2.5}$, PM_{10}, NO2, NOx at residence from land-use regression	United Kingdom (UK Biobank)

Provenance and method

Period **4 – 10 October 2026**, Sunday to Saturday, **seven daily issues**, **87 records**, **87 unique DOIs**, no record without a DOI. This rollup is **aggregation only**: it re-reads `state/corpus/*.json`, `state/metrics.csv` and `state/trials.json` over the period and **queries no literature API**; every record was gated by `check_dois.py` at daily-issue time. Tier mix: **18 A, 44 B, 25 C** — the highest tier-A share of the series (21 %), on its smallest week. The 4–8 October dailies were built on the **9 October recovery run**; the 9 and 10 October dailies and this rollup were built on the **12 October run**, which also shipped the 11 October daily and closed the backlog. Four records admitted to a daily issue in or before this period are **held, not rejected** and are absent from this pooled corpus: three from 9 October (interrupted authoring) and one from 3 October. Effect estimates are transcribed verbatim from the daily issues on heterogeneous contrasts and are **not** meta-analysed. Abstracts remain the copyright of their respective publishers.